Robot motion

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Reference Books:

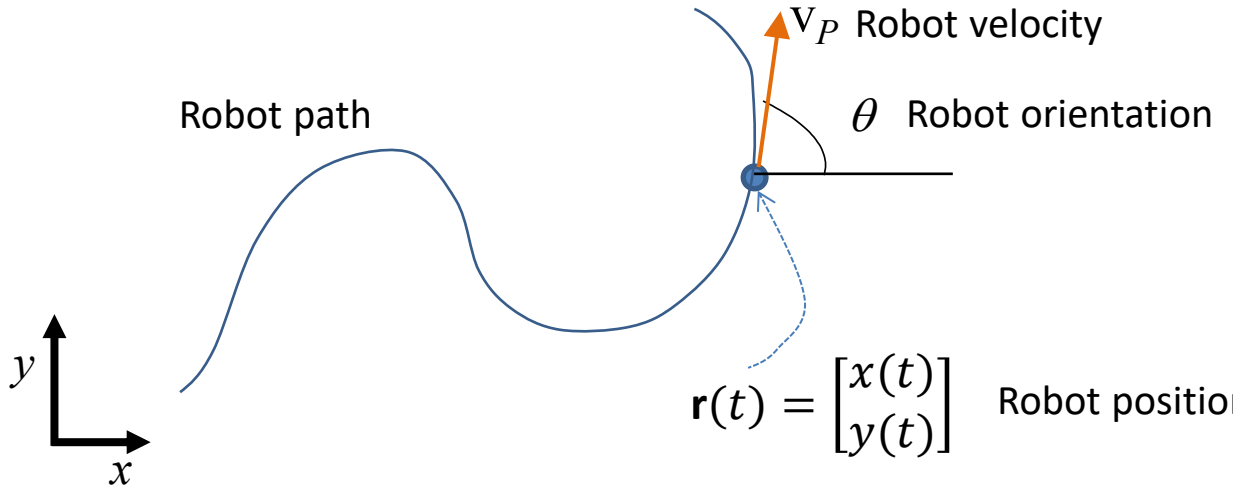
- Probabilistic Robotics. S. Thrun, W. Burgard, D. Fox. MIT Press. 2005.
- Simultaneous Localization and Mapping for Mobile Robots: Introduction and Methods. Juan-Antonio Fernández-Madrigal and José Luis Blanco Claraco. IGI-Global. 2013.

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Wheeled-Robot Kinematics

Locomotion: The act of moving from one place to another



Robot pose $\mathbf{x}$: position ($\mathbf{r}$) and orientation (θ)

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} \mathbf{r} \\ \theta \end{bmatrix}$$

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$ Robot position at time t in a given coordinate system

- A vehicle is holonomic if it can move in any arbitrary direction.
- The number of its **local** *d*egrees-*of*-*f*ree*dom* (dof) of movement equals the number of **global** dof (3 for planar motion, 6 in 3D space)
- A **degree of freedom (dof)** refers to an **independent way in which a robot can move**

A non-holonomic wheeled robot is subject to motion constraints → the lateral velocity is zero (the robot **cannot move sideways** instantaneously).

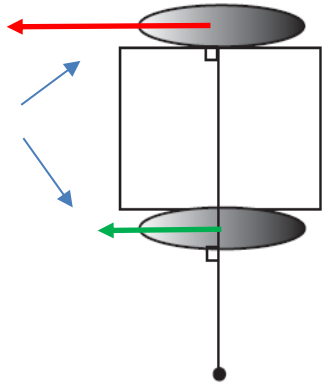
Wheeled-Robot Kinematics

Two main motion configurations for non-holonomic wheeled robots

We'll focus on this configuration

Differential drive

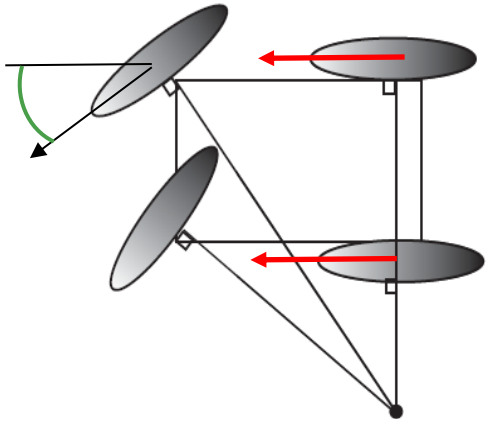
2 actuators: velocities of the 2 wheels



ICR

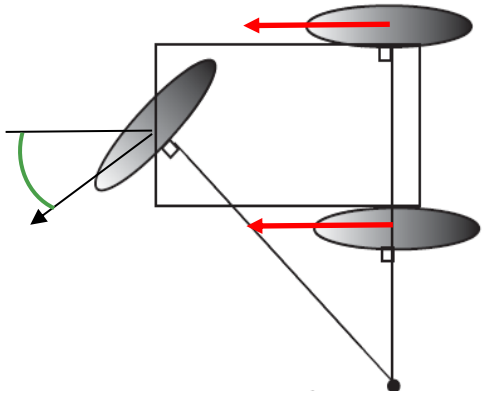
Ackermann steering

Car drive



ICR

Bicycle



ICR

2 actuators: velocity of the rear wheels, and steering of the front wheel

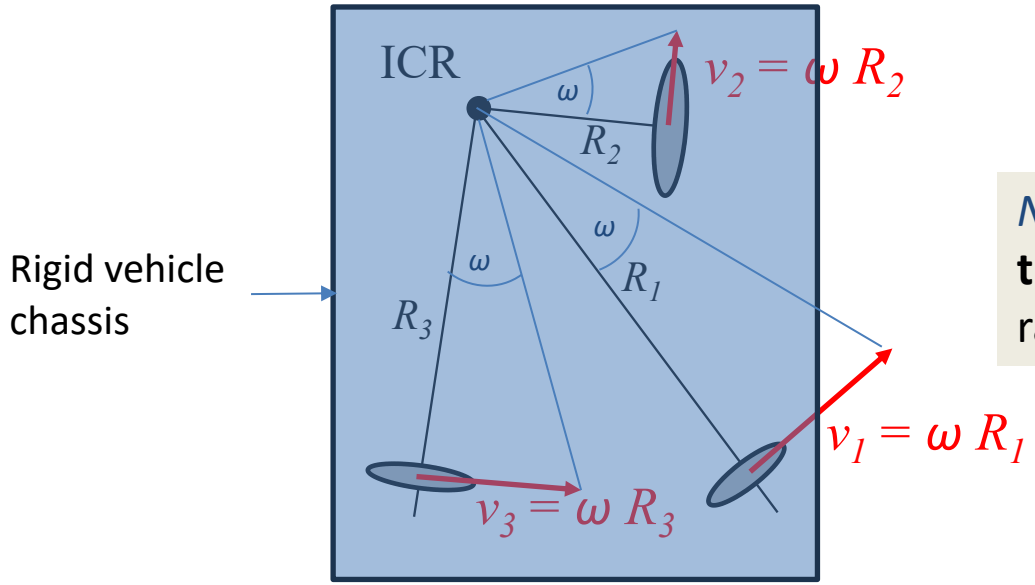
The ICR (Instantaneous Center of Rotation) is the intersection of the extended wheel axes.

All these are **Non-holonomic vehicles**:

- 3 pose components: $\mathbf{x} = [x, y, \theta]^T$
- 2 independent control variables (actuations)

Wheeled-Robot Kinematics

Rigid-body motion about the ICR



Note: **ω** is the vehicle's **yaw rate about the ICR**, not the individual wheel-spin rates ω_l, ω_r (to be introduced later)

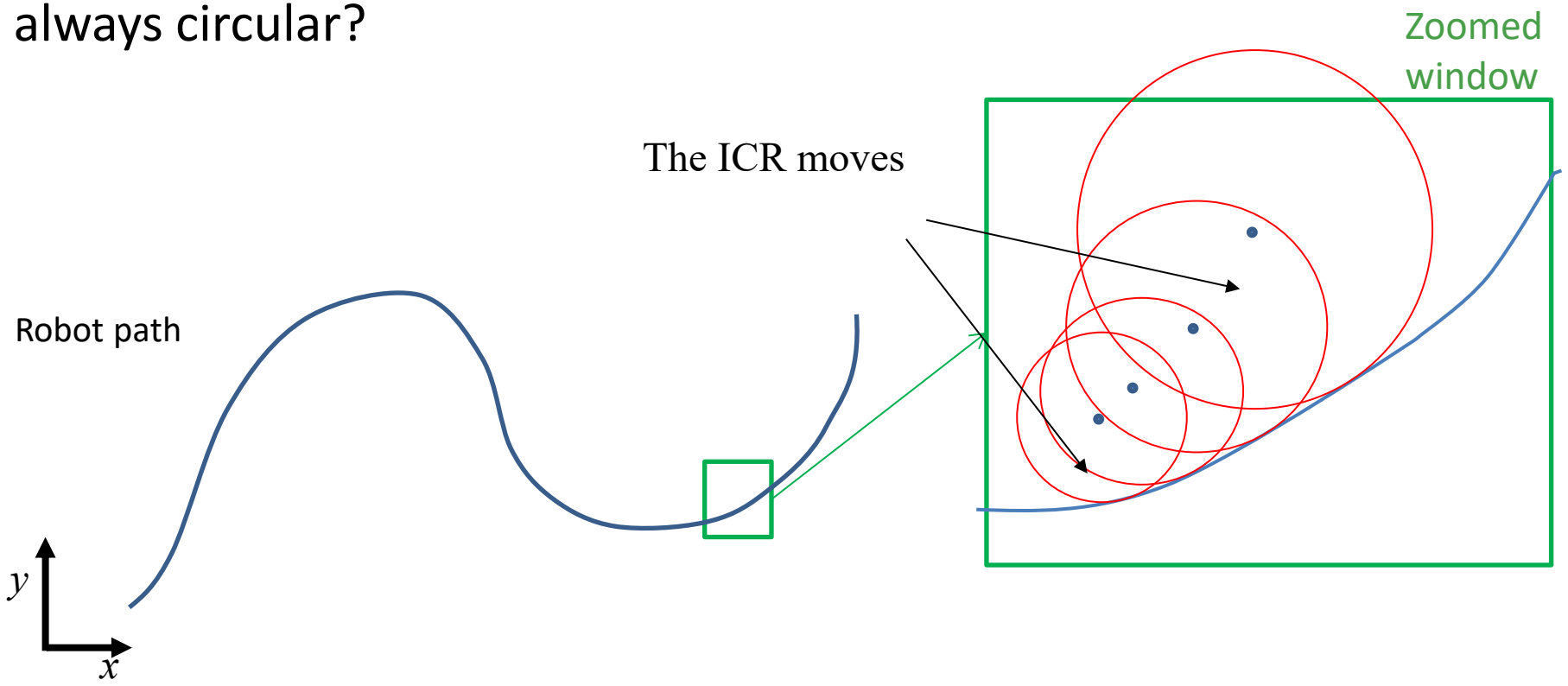
Just one ICR and 3 Wheels: the vehicle will move by rolling the 3 wheels (no slip) rotating around the ICR with angular velocity ω (yaw rate).

- A rigid chassis has only one ICR and one ω (otherwise there will be slip of the wheels)
- Wheel i follows a circle of radius R_i
- Its wheel-center speed is: $v_i = \omega R_i$

Wheeled-Robot Kinematics

Differential Drive

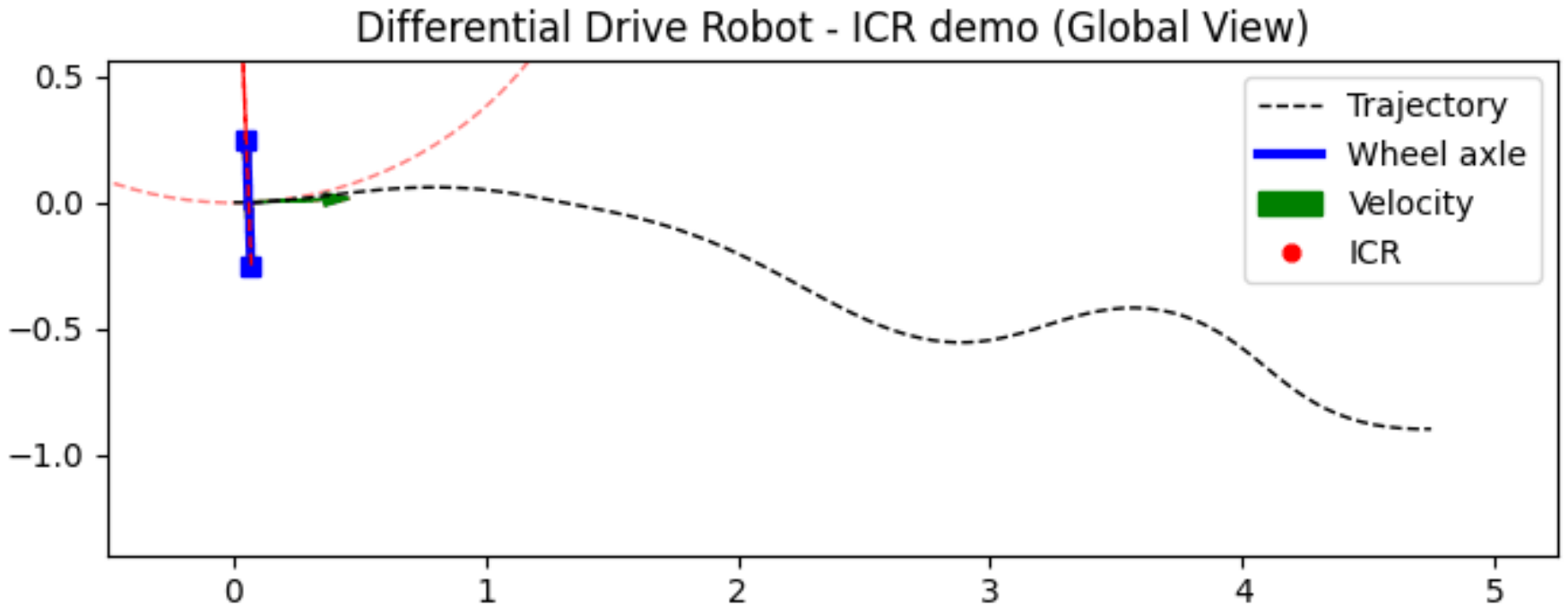
How is it possible that a robot moves along this path if the instantaneous motion is always circular?



The path is the composition of a sequence of circular motions with an ICR that moves

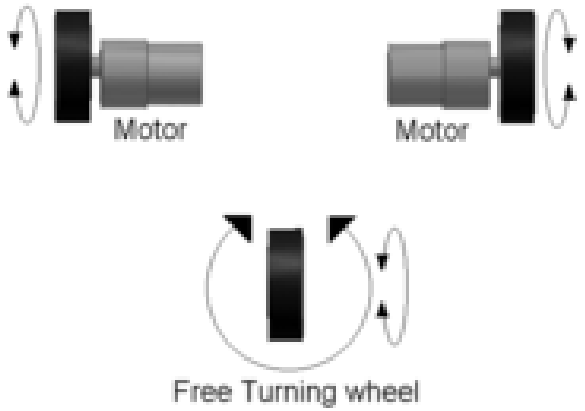
Wheeled-Robot Kinematics

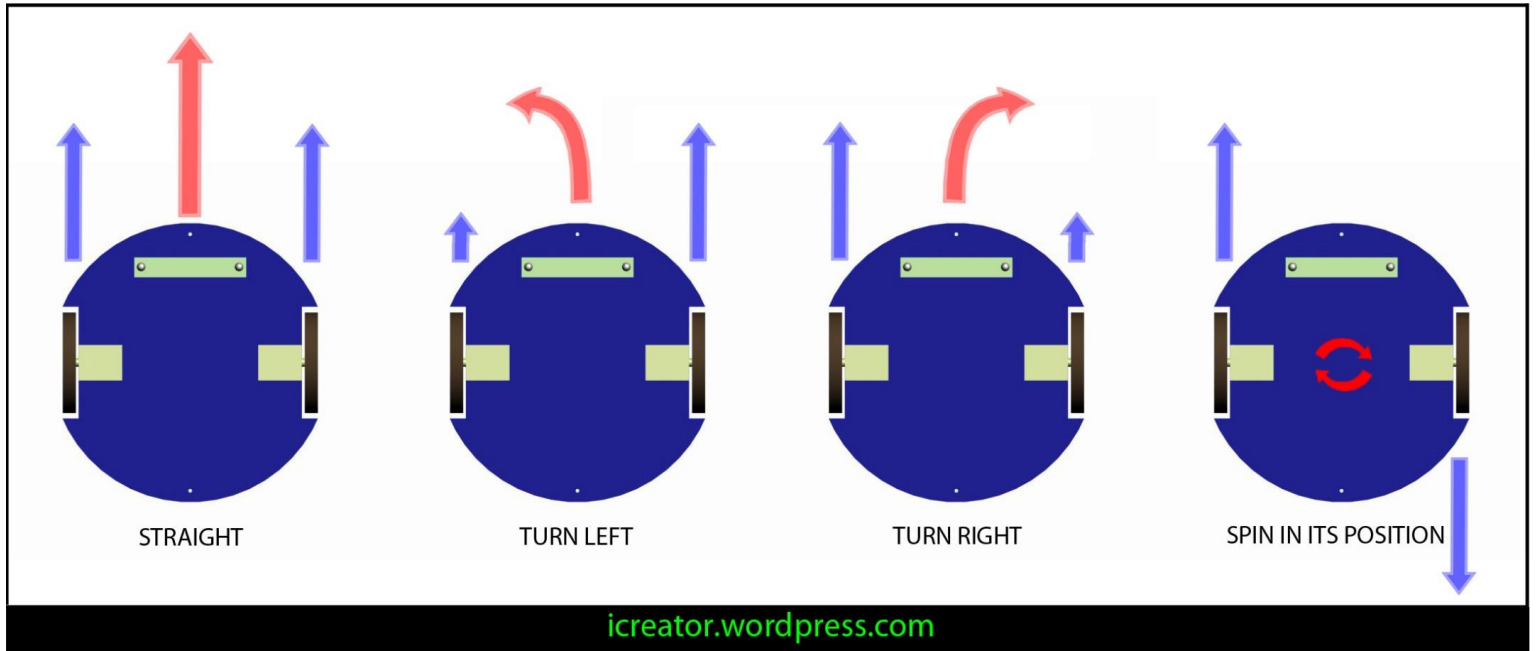
Differential Drive



Wheeled-Robot Kinematics

Differential Drive

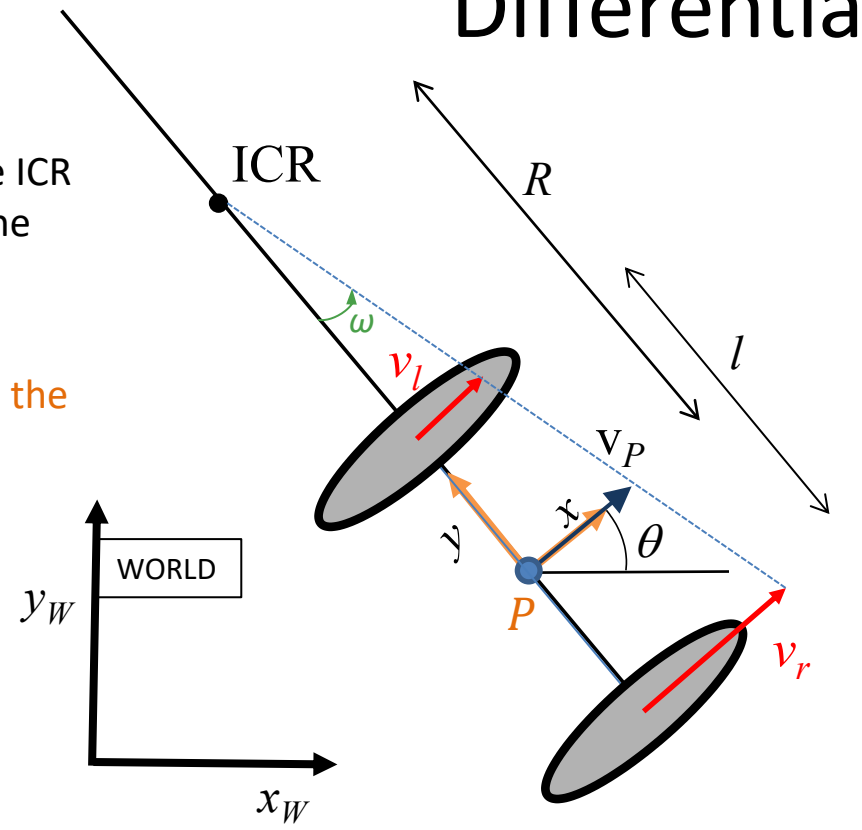




Differential Drive

- ICR somewhere along this line
- v_l and v_l gives the ICR position in the line

Robot frame located at the wheel axle midpoint *P*



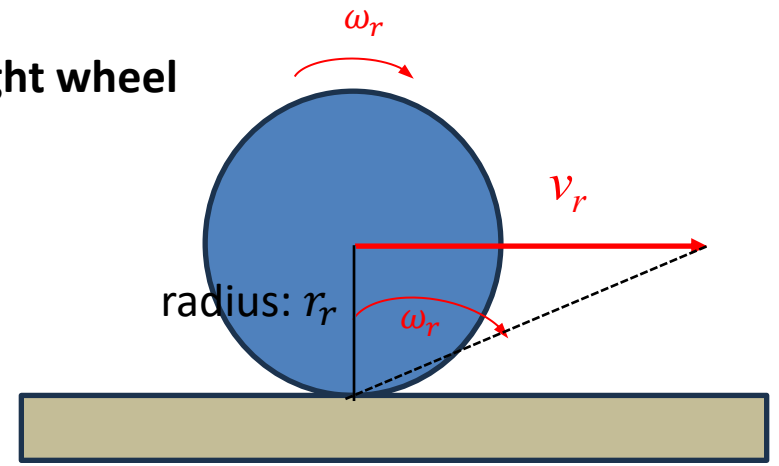
From the wheel-center velocities of the wheels v_l, v_r:

$$R = \frac{l}{2}\frac{(v_l + v_r)}{(v_r - v_l)} \quad \omega = \frac{v_r - v_l}{l}$$

Linear velocity of the **axle midpoint** P : $v_P = \omega \cdot R$

$$v_P = \frac{v_l + v_r}{2}$$

Right wheel



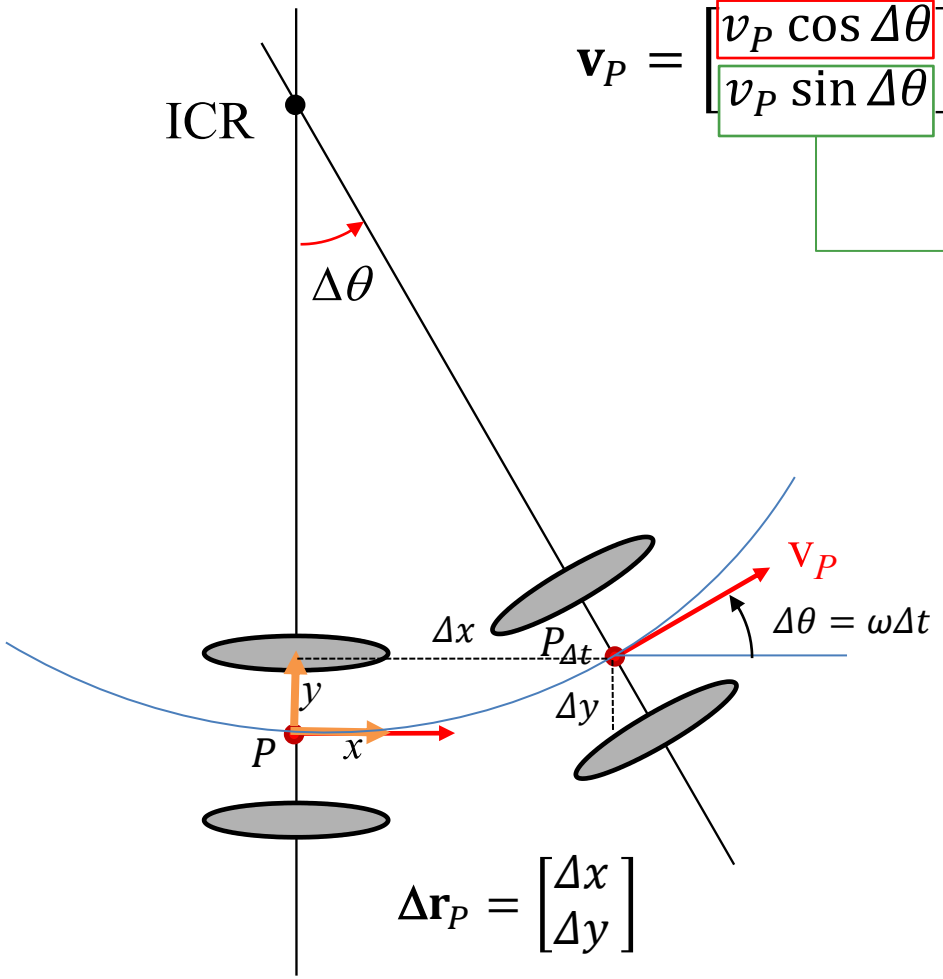
Angular velocities of the wheels: $\omega_r = \frac{v_r}{r_r} \quad \omega_l = \frac{v_l}{r_l}$

To follow a path, the motion command can be either $\langle \omega_l, \omega_r \rangle$ or $\langle v_P, \omega \rangle$

$$\langle \omega_l, \omega_r \rangle \rightarrow \langle v_l, v_r \rangle \rightarrow \langle v_P, \omega \rangle$$

Differential Drive

Incremental robot pose: How much has the robot moved during Δt



$$\mathbf{v}_P = \begin{bmatrix} v_P \cos\Delta\theta \\ v_P \sin\Delta\theta \end{bmatrix}$$

$$\Delta x = \int_0^{\Delta t} v_P(t)\ \cos[\Delta\theta(t)]dt = \frac{v_P}{\omega}\sin(\omega\Delta t) = R\sin(\omega\Delta t)$$

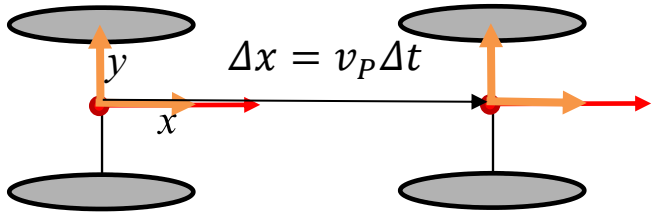
$$\Delta y = \int_0^{\Delta t} v_P(t)\ \sin[\Delta\theta(t)]dt = \frac{v_P}{\omega}[1-\cos(\omega\Delta t)] = R[1-\cos(\omega\Delta t)]$$

$$\Delta\theta = \int_0^{\Delta t} \omega(t)\ dt = \omega\Delta t$$

$\Delta\theta = \omega\Delta t$

Assuming ω and R constant during the period Δt

If the robot moves *straight (no rotation)* $\omega = 0 \rightarrow v_r = v_l = v_P$



ICR at infinite

$$\Delta x = v_P\Delta t$$
$$\Delta y = 0$$
$$\Delta\theta = 0$$

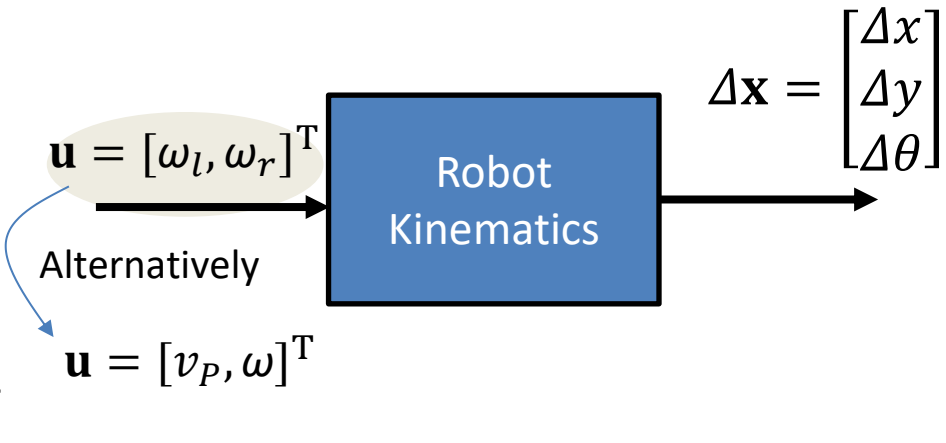
Wheeled-Robot Kinematics

Summary of Differential Drive:

$$v_r = \omega_r r_r$$
$$v_l = \omega_l r_l$$

$$\omega = \frac{v_r - v_l}{l} = \frac{r_r\omega_r - r_l\omega_l}{l}$$

$$v_P = \frac{v_l + v_r}{2} = \frac{r_r\omega_r + r_r\omega_r}{2}$$



Robot pose expressed in the current robot (local) coordinate system!

$$\Delta x = \frac{v_P}{\omega}\sin(\omega\Delta t)$$
$$\Delta y = \frac{v_P}{\omega}[1 - \cos(\omega\Delta t)]$$
$$\Delta\theta = \omega\Delta t$$

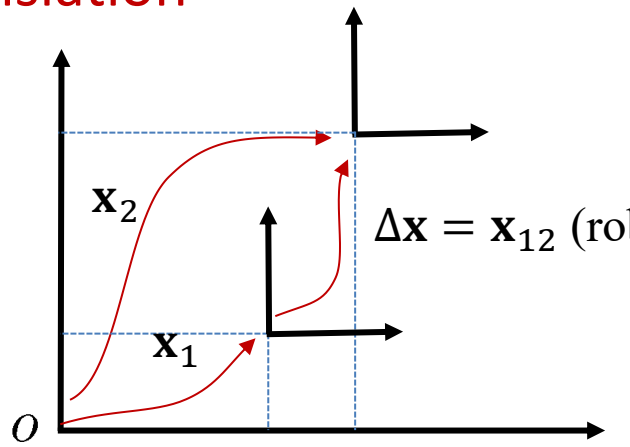
What if we want the path traversed by the robot?

Pose increments must be composed with the previous pose in a global coordinate system.

Planar poses and transformations

What is the pose $\mathbf{x}_2$ of a robot if it moved $\Delta\mathbf{x} = \mathbf{x}_{12}$ from pose $\mathbf{x}_1$?

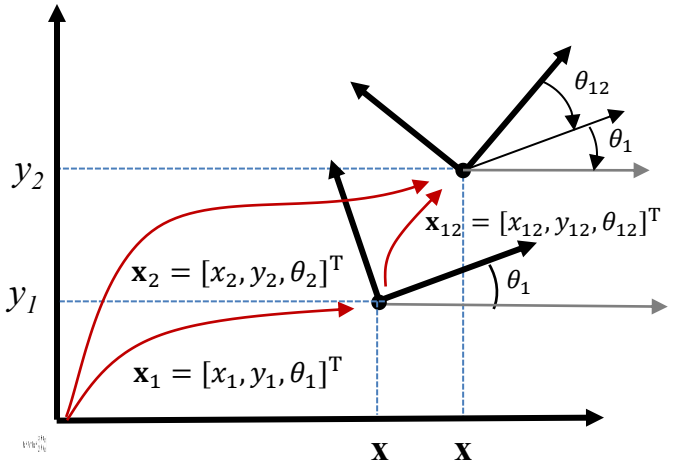
Just a translation



$$\mathbf{x}_2 = \mathbf{x}_1 + \mathbf{x}_{12} = \begin{bmatrix} x_1 + x_{12} \\ y_1 + y_{12} \end{bmatrix}$$

Behave like vectors!

Translation + rotation: $\mathbf{x}_2 \neq \mathbf{x}_1 + \mathbf{x}_{12}$ → $\mathbf{x}_2 = \mathbf{x}_1 \oplus \mathbf{x}_{12}$ **Pose composition**

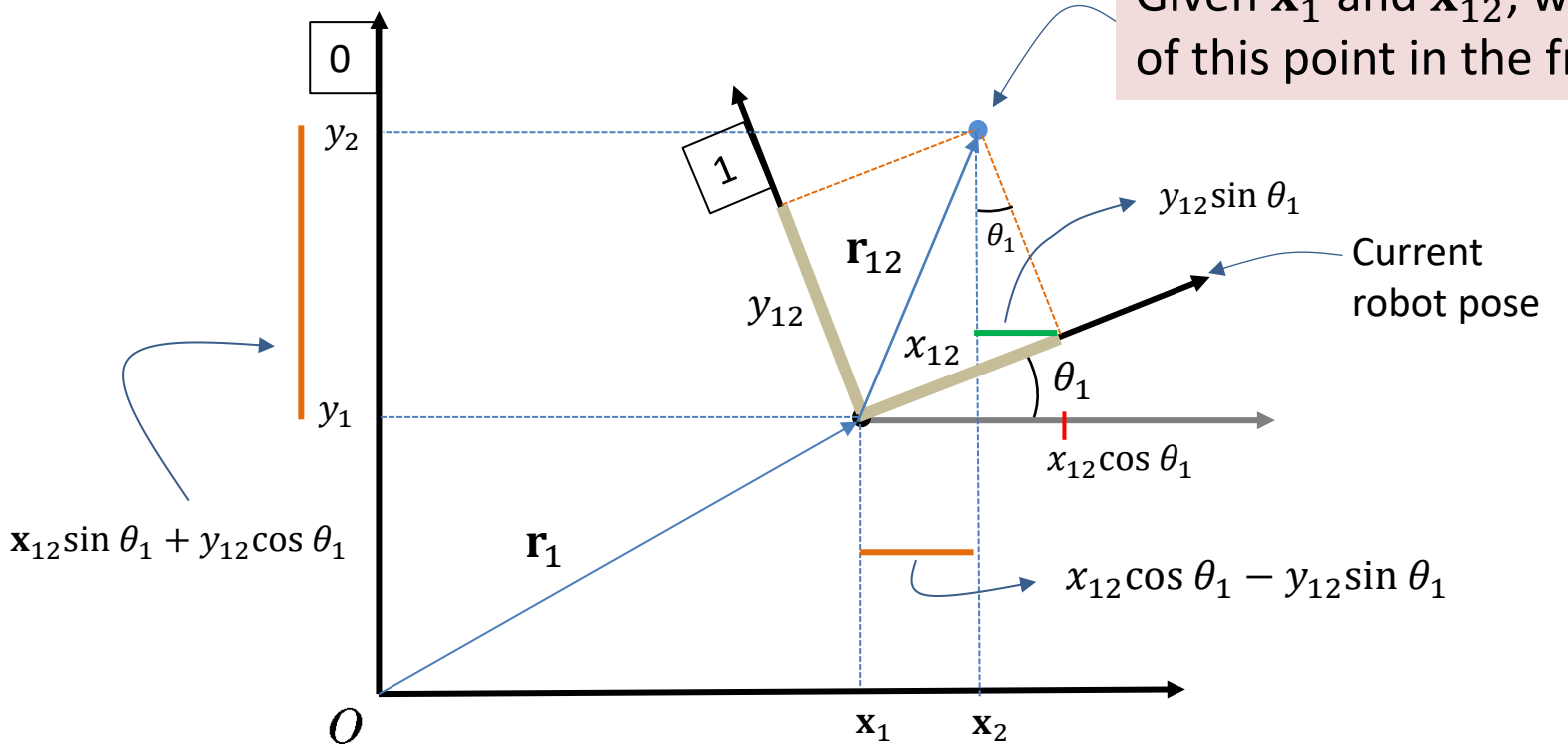


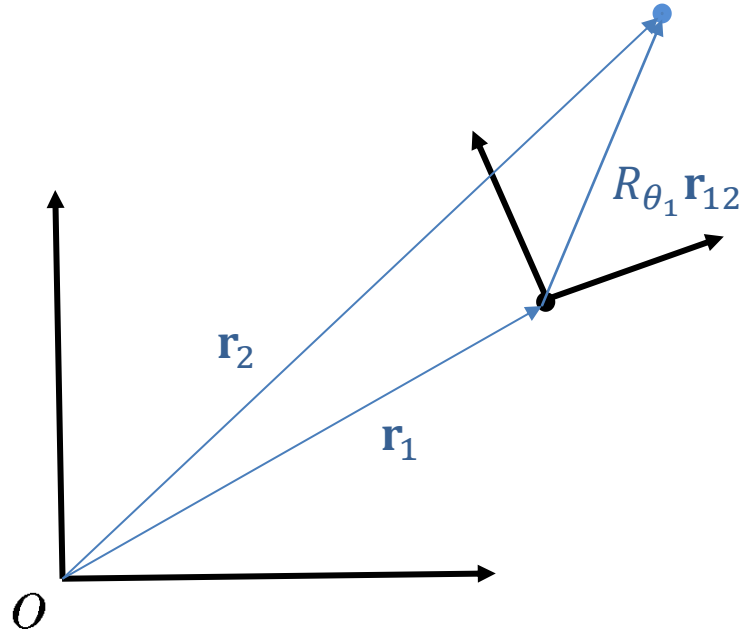
$$\mathbf{x}_2 = \begin{bmatrix} x_2 \\ y_2 \\ \theta_2 \end{bmatrix} = \mathbf{x}_1 \oplus \mathbf{x}_{12} = \begin{bmatrix} x_1 + x_{12}\cos\theta_1 - y_{12}\sin\theta_1 \\ y_1 + x_{12}\sin\theta_1 + y_{12}\cos\theta_1 \\ \theta_1 + \theta_{12} \end{bmatrix}$$

Where does this expression come from? NEXT

Rigid (Euclidean) transformation between coordinate systems

Given $\mathbf{x}_1$ and $\mathbf{x}_{12}$, what are the coordinates of this point in the frame "0"?



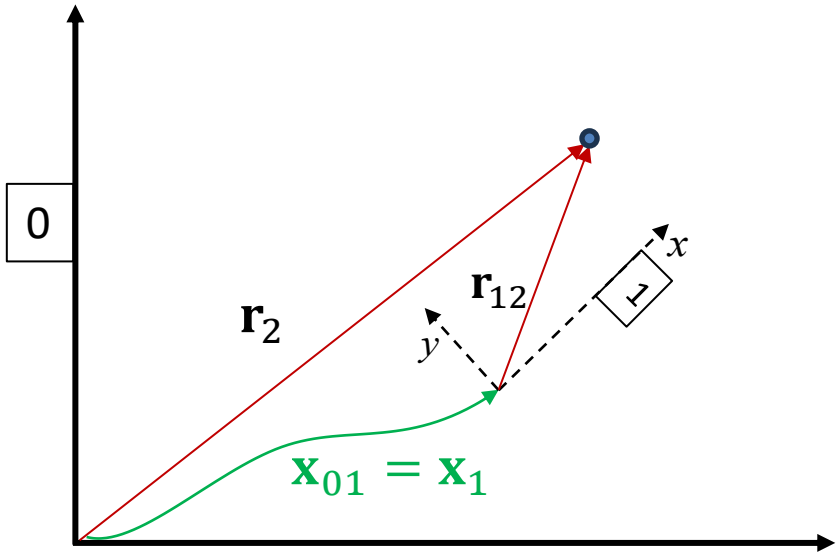


This is the matrix to remember

$$\mathbf{r}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_{12}\cos\theta_1 & -y_{12}\sin\theta_1 \\ x_{12}\sin\theta_1 & y_{12}\cos\theta_1 \end{bmatrix} + \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix} \begin{bmatrix} x_{12} \\ y_{12} \end{bmatrix} + \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = R_{\theta_1}\mathbf{r}_{12} + \mathbf{r}_1$$

If no rotation $R_{\theta_1} = I$ ⟹ $\mathbf{r}_2 = \mathbf{r}_{12} + \mathbf{r}_1$

Pose as a Euclidean transformation between coordinate systems



$$\mathbf{r}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = R_{\theta_1}\mathbf{r}_{12} + \mathbf{r}_1$$

Using homogeneous coordinates: $\tilde{\mathbf{r}}_2 = \begin{bmatrix} \mathbf{r}_2 \\ 1 \end{bmatrix}$

$$\tilde{\mathbf{r}}_2 = \begin{bmatrix} \mathbf{r}_2 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R_{\theta_1} & \mathbf{r}_1 \\ 0^{\mathrm{T}} & 1 \end{bmatrix}}_{{}^0T_1} \underbrace{\begin{bmatrix} \mathbf{r}_{12} \\ 1 \end{bmatrix}}_{\tilde{\mathbf{r}}_{12}} = {}^0T_1\tilde{\mathbf{r}}_{12}$$

${}^0T_1$ expresses point coordinates given in the **system 1** as coordinates of **system 0**

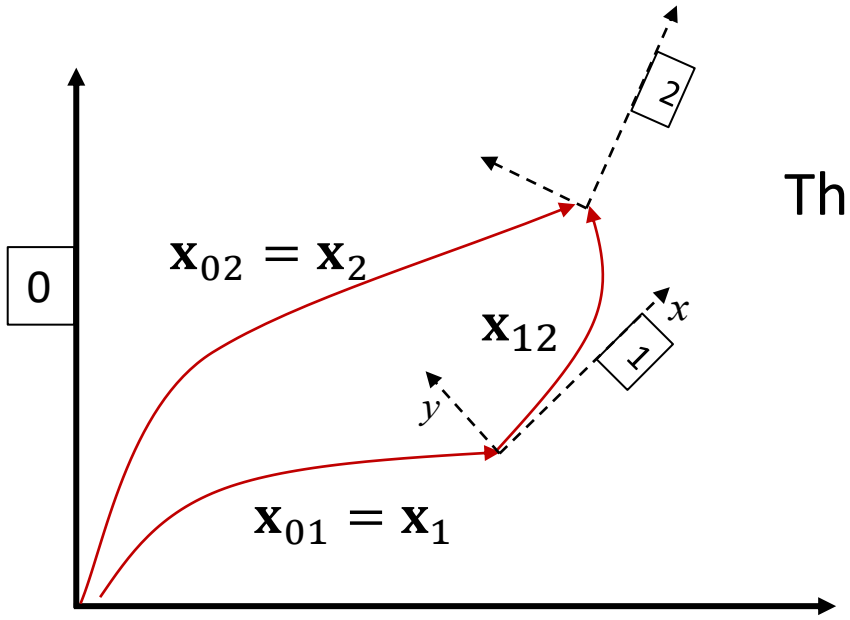
Equivalence between transformation and pose (${}^0T_1 = T(\mathbf{x}_{01})$):

$$\mathbf{x}_{01} = \mathbf{x}_1 = \begin{bmatrix} x_1 \\ y_1 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \\ \theta_1 \end{bmatrix} \quad \rightarrow \quad T(\mathbf{x}_{01}) = {}^0T_1 = \begin{bmatrix} R_{\theta_1} & \mathbf{r}_1 \\ 0^{\mathrm{T}} & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & x_1 \\ \sin\theta_1 & \cos\theta_1 & y_1 \\ 0 & 0 & 1 \end{bmatrix}$$

pose 1 wrt frame 0 Typically, 0 is dropped for short

The vector $\mathbf{x}_1$ and matrix ${}^0T_1$ represent the same pose, but they are different mathematical objects.

Planar poses and transformations



The robot, at pose $\mathbf{x}_1$, moves $\mathbf{x}_{12}$ reaching a new pose $\mathbf{x}_2$

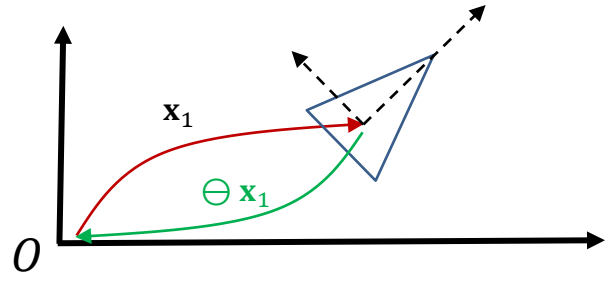
$$\mathbf{x}_2 = \mathbf{x}_1 \oplus \mathbf{x}_{12}$$

$$\mathbf{x}_2 = \begin{bmatrix} x_2 \\ y_2 \\ \theta_2 \end{bmatrix} = \mathbf{x}_1 \oplus \mathbf{x}_{12} = \begin{bmatrix} x_1 + x_{12}\cos\theta_1 - y_{12}\sin\theta_1 \\ y_1 + x_{12}\sin\theta_1 + y_{12}\cos\theta_1 \\ \theta_1 + \theta_{12} \end{bmatrix} = \begin{bmatrix} {}^0T_1\tilde{\mathbf{r}}_{12}(1) \\ {}^0T_1\tilde{\mathbf{r}}_{12}(2) \\ \theta_1 + \theta_{12} \end{bmatrix}$$

First two elements of the 3x1 vector ${}^0T_1\tilde{\mathbf{r}}_{12}$

Why is it useful to represent poses as transformations?

1. **Inverse of a pose** (inverse of transformation)



$$^0T_1 = T(\mathbf{x}_1) \Rightarrow \boxed{^1T_0 = (\,^0T_1)^{-1} = T(\ominus\,\mathbf{x}_1)}$$

RECALL: $$^1T_0 = (\,^0T_1)^{-1} = \begin{bmatrix} R_{\theta_1} & \mathbf{r}_1 \\ 0^{\mathrm{T}} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R_{\theta_1}^{-1} & -R_{\theta_1}^{-1}\mathbf{r}_1 \\ 0^{\mathrm{T}} & 1 \end{bmatrix} = \boxed{\begin{bmatrix} R_{\theta_1}^{\mathrm{T}} & -R_{\theta_1}^{\mathrm{T}}\mathbf{r}_1 \\ 0^{\mathrm{T}} & 1 \end{bmatrix}}$$

$^1T_0$ transforms the coordinates of a point expressed in frame 0 into coordinates expressed in frame 1

Example: $\mathbf{r}_O = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ origin O expressed in the system 0

$$^1\tilde{\mathbf{r}}_{O_0} = {}^1T_0\ {}^0\tilde{\mathbf{r}}_{O_0} = \begin{bmatrix} R_{\theta_1}^{\mathrm{T}} & -R_{\theta_1}^{\mathrm{T}}\mathbf{r}_1 \\ 0^{\mathrm{T}} & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} c\theta_1 & s\theta_1 \\ -s\theta_1 & c\theta_1 \end{bmatrix} & -\begin{bmatrix} x_1c\theta_1 + y_1s\theta_1 \\ -x_1s\theta_1 + y_1c\theta_1 \end{bmatrix} \\ [0 \quad 0] & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -x_1c\theta_1 - y_1s\theta_1 \\ x_1c\theta_1 - y_1s\theta_1 \\ 1 \end{bmatrix}$$

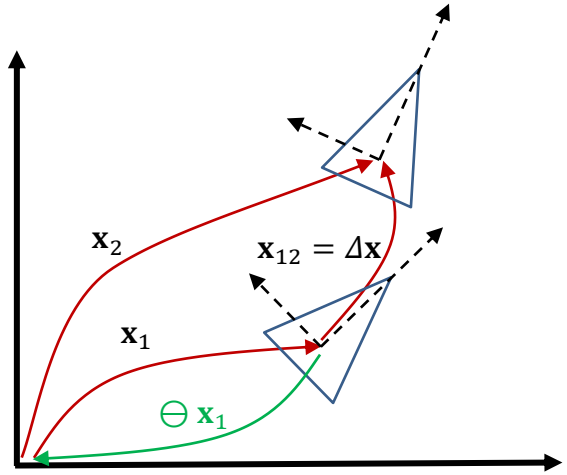
Example: We know the pose of a robot B *wrt* to a robot A is $\mathbf{x}_{AB} = [2\ 3\ 45^{\mathrm{o}}]^{\mathrm{T}}$

What is the pose of the robot A *wrt* to robot B ?

Answer: $\mathbf{x}_{BA} = \ominus\,\mathbf{x}_{AB} = \left[\frac{-5\sqrt{2}}{2} \quad -\frac{\sqrt{2}}{2} \quad -45^{\mathrm{o}}\right]^{\mathrm{T}}$

1. Inverse of a pose. Use case: computation of $\Delta\mathbf{x}$

Do not memorize these equations!



Composition

$$\mathbf{x}_2 = \begin{bmatrix} x_2 \\ y_2 \\ \theta_2 \end{bmatrix} = \mathbf{x}_1 \oplus \Delta\mathbf{x} = \begin{bmatrix} x_1 + \Delta x \cos\theta_1 - \Delta y \sin\theta_1 \\ y_1 + \Delta x \sin\theta_1 + \Delta y \cos\theta_1 \\ \theta_1 + \Delta\theta \end{bmatrix}$$

Inverse

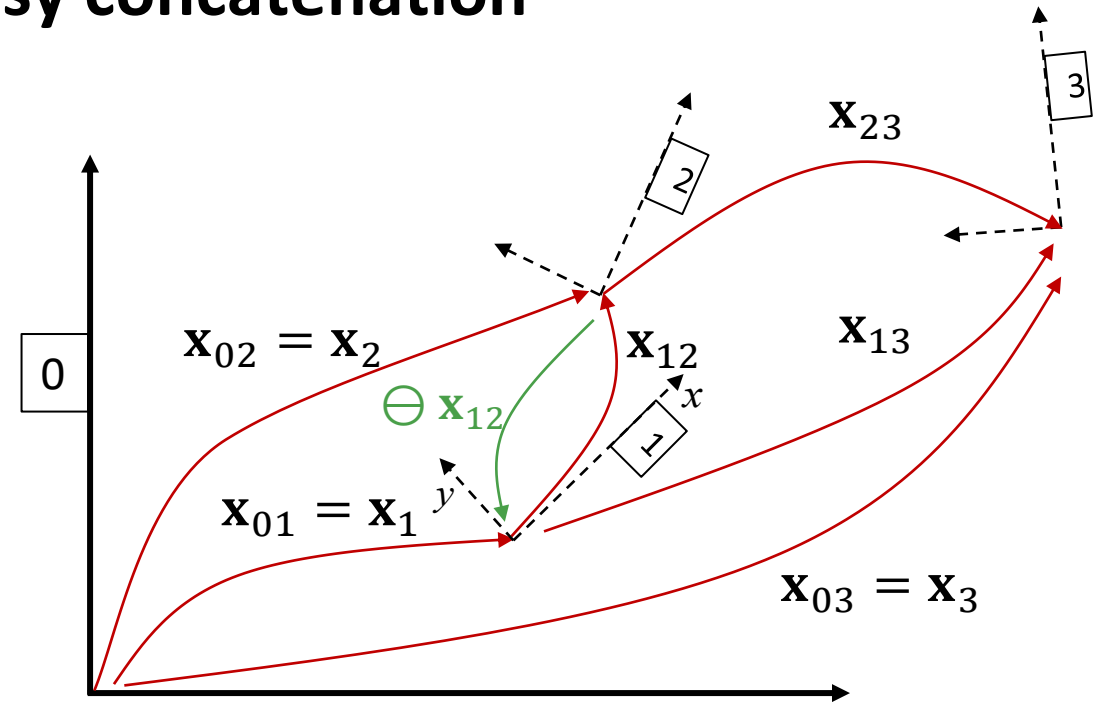
$$\ominus\, \mathbf{x}_1 = \begin{bmatrix} x_1^- \\ y_1^- \\ \theta_1^- \end{bmatrix} = \begin{bmatrix} -R_{\theta_1}^{\mathrm{T}} r_1 \\ -\theta_1 \end{bmatrix} = \begin{bmatrix} -x_1 \cos\theta_1 - y_1 \sin\theta_1 \\ x_1 \sin\theta_1 - y_1 \cos\theta_1 \\ -\theta_1 \end{bmatrix}$$

$$\mathbf{x}_2 = \mathbf{x}_1 \oplus \Delta\mathbf{x}$$

$$\Delta\mathbf{x} = (\ominus\, \mathbf{x}_1) \oplus \mathbf{x}_2 = \begin{bmatrix} x_1^- + x_2 \cos\theta_1^- - y_2 \sin\theta_1^- \\ y_1^- + x_2 \sin\theta_1^- + y_2 \cos\theta_1^- \\ \theta_1^- + \theta_2 \end{bmatrix} = \begin{bmatrix} (x_2 - x_1)\cos\theta_1 + (y_2 - y_1)\sin\theta_1 \\ -(x_2 - x_1)\sin\theta_1 + (y_2 - y_1)\cos\theta_1 \\ \theta_2 - \theta_1 \end{bmatrix}$$

Why is it useful to represent poses as transformations?

2. Easy concatenation



Sequence of transformation: $\mathbf{x}_{02} = \mathbf{x}_{01} \oplus \mathbf{x}_{12} \longrightarrow {}^{0}T_2 = {}^{0}T_1 \, {}^{1}T_2$

Solve for any pose: $\mathbf{x}_{21} = \ominus \mathbf{x}_{12} = \mathbf{x}_{23} \oplus (\ominus \mathbf{x}_{13}) \longrightarrow {}^{1}T_2^{-1} = {}^{2}T_1 = {}^{2}T_3 \, {}^{3}T_1 = {}^{2}T_3 \, {}^{1}T_3^{-1}$

$\mathbf{x}_2 = \mathbf{x}_1 \oplus \mathbf{x}_{13} \oplus (\ominus \mathbf{x}_{23}) \longrightarrow {}^{0}T_2 = {}^{0}T_1 \, {}^{1}T_3 \, {}^{3}T_2$

Properties of poses

The set of 2D poses equipped with the composition operator $\oplus$ forms the noncommutative group $SE(2)$ (Special Euclidean).

Closure: $\mathbf{x}_A \oplus \mathbf{x}_B = \mathbf{x}_C$ The composition of poses gives us a pose

Associativity: $(\mathbf{x}_A \oplus \mathbf{x}_B) \oplus \mathbf{x}_C = \mathbf{x}_A \oplus (\mathbf{x}_B \oplus \mathbf{x}_C)$

Identity element: $\mathbf{x}_A \oplus 0 = 0 \oplus \mathbf{x}_A = \mathbf{x}_A$

Inverse: $\ominus \mathbf{x}_{AB} = \mathbf{x}_{BA}$ $\quad (\ominus \mathbf{x}_A) \oplus \mathbf{x}_A = 0, \quad \mathbf{x}_A \oplus (\ominus \mathbf{x}_A) = 0$

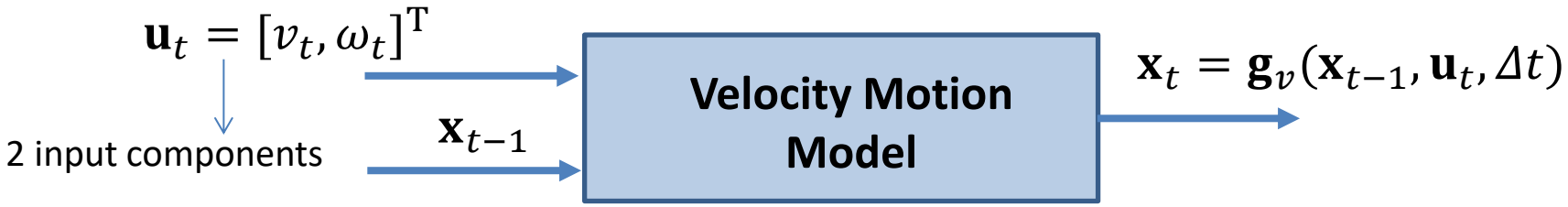
Watch out: Not commutative $\mathbf{x}_A \oplus \mathbf{x}_B \neq \mathbf{x}_B \oplus \mathbf{x}_A$

Deterministic Motion Models

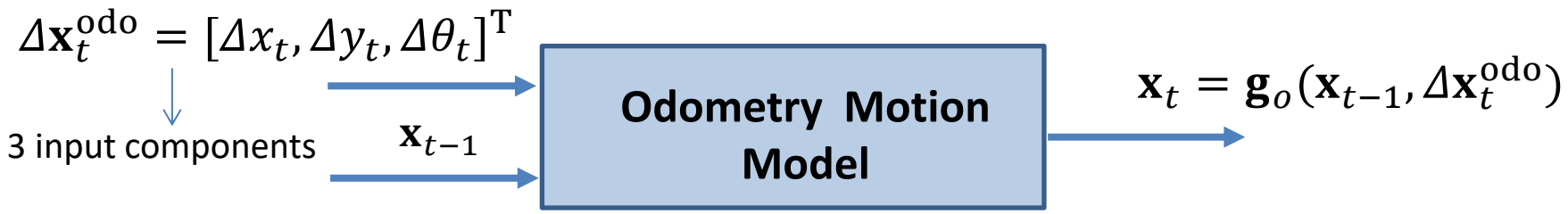
Motion model: predicts the robot pose at time t from its previous pose and the motion input applied over $[t-1, t]$.

Two formulations, depending on the input $\mathbf{u}_t$:

- **Velocity Motion Model**: the input $\mathbf{u}_t$ is a **velocity vector** ($\mathbf{u}_t = [v_t, \omega_t]^{\mathrm{T}}$)



- **Odometry Motion Model:** the input $\Delta\mathbf{x}_t^{\mathrm{odo}}$ is a **pose increment** given by the odometry readings at time $t-1$ and t ($\Delta\mathbf{x}_t^{\mathrm{odo}} = (\ominus\, \mathbf{x}_{t-1}^{\mathrm{odo}}) \oplus \mathbf{x}_t^{\mathrm{odo}}$)



Deterministic Motion Models

Velocity-input Motion Model:

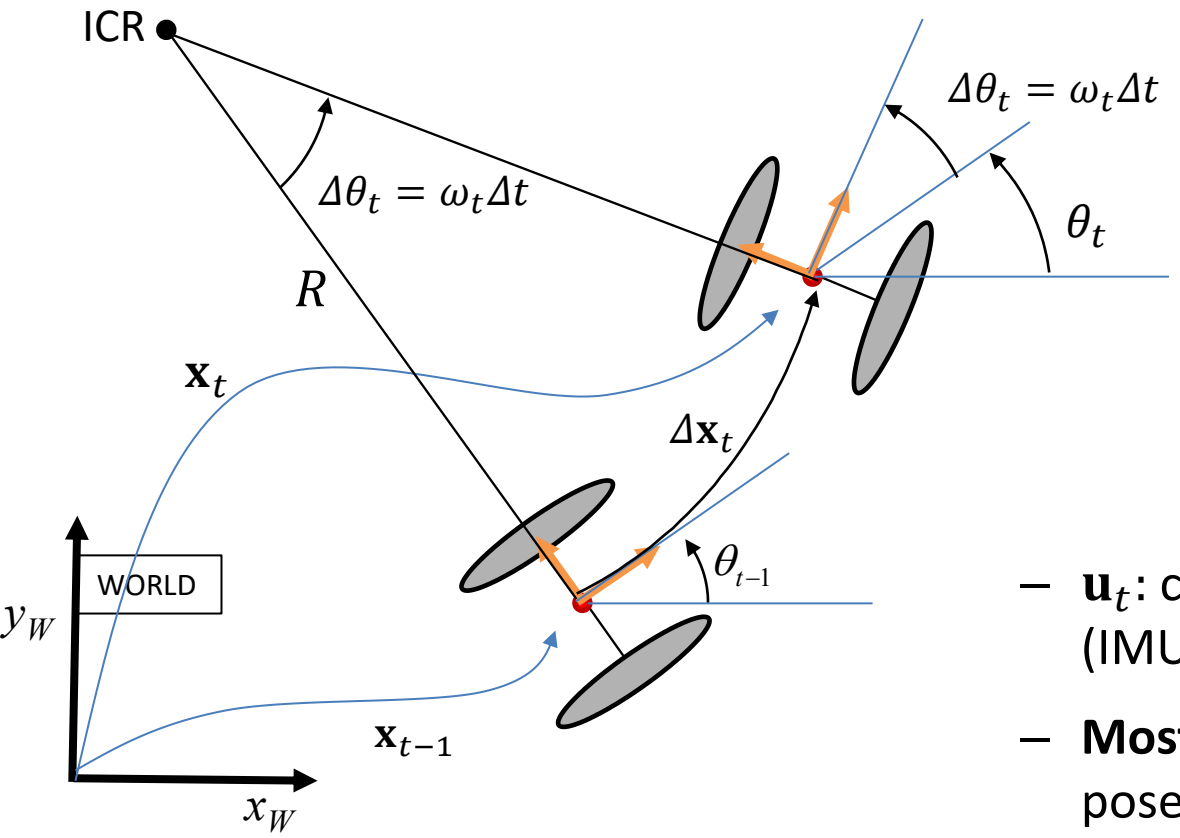
ω_t and v_t are constant over Δt

$\mathbf{u}_t = [v_t, \omega_t]^{\mathrm{T}}$ (2 variables)

$\mathbf{x}_{t-1}$

Velocity Motion Model

$$\mathbf{x}_t = \mathbf{g}_v(\mathbf{x}_{t-1}, \mathbf{u}_t, \Delta t) = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t(\mathbf{u}_t, \Delta t)$$



$$\mathbf{x}_t = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t = \begin{bmatrix} x_{t-1} + \Delta x_t \cos\theta_{t-1} - \Delta y_t \sin\theta_{t-1} \\ y_{t-1} + \Delta x_t \sin\theta_{t-1} + \Delta y_t \cos\theta_{t-1} \\ \theta_{t-1} + \Delta\theta_t \end{bmatrix}$$

If $\omega \neq 0$

$$\Delta x_t = \frac{v_t}{\omega_t}\sin(\omega_t \Delta t)$$
$$\Delta y_t = \frac{v_t}{\omega_t}[1 - \cos(\omega_t \Delta t)]$$
$$\Delta\theta_t = \omega_t \Delta t$$

If straight-line motion ($\omega = 0$)

$$\Delta x_t = v_t \Delta t$$
$$\Delta y_t = 0$$
$$\Delta\theta_t = 0$$

- $\mathbf{u}_t$: command sent to the robot or estimated from sensor data (IMU, visual data, …)
- **Mostly used for:** trajectory tracking and planning, simulation, pose prediction (in EKF), …

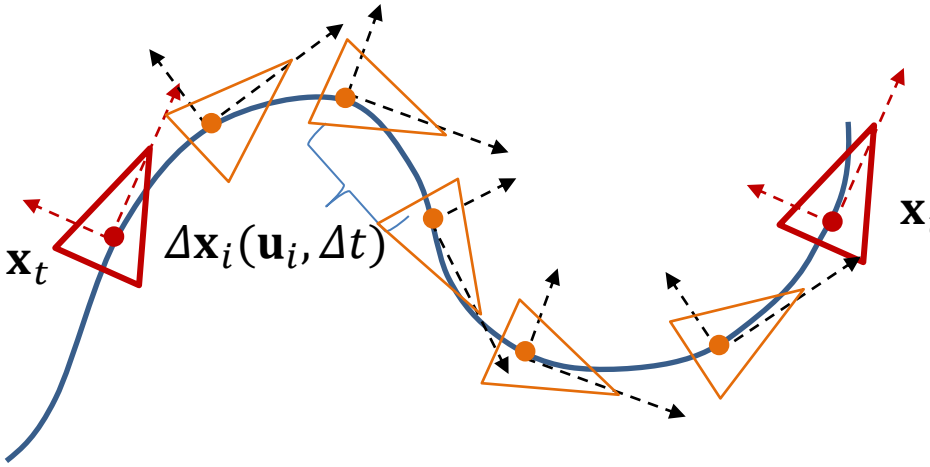
Deterministic Motion Models

Velocity-input Motion Model:

ω_t and v_t are constant over Δt



A sequence of velocity inputs $\{\mathbf{u}_i\}_{i=1}^{N}$, with each input held constant over Δt, generates a trajectory composed of constant-curvature segments.



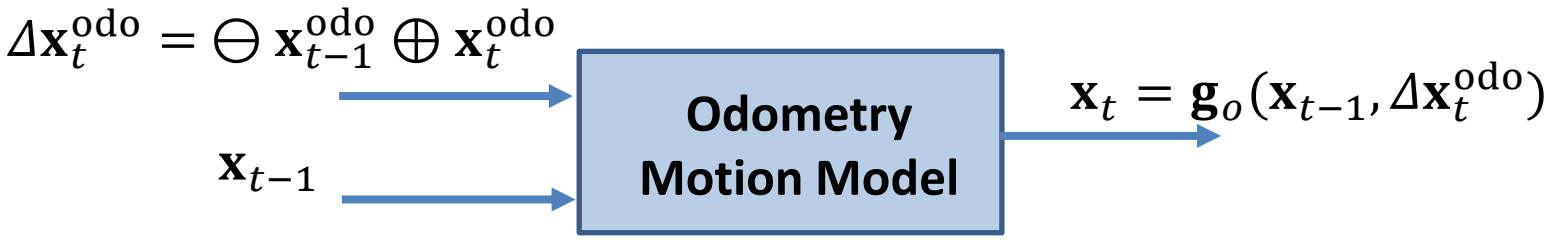
$$\mathbf{x}_{t+N} = \mathbf{x}_t \oplus \Delta\mathbf{x}_1(\mathbf{u}_1, \Delta t) \oplus \cdots \oplus \Delta\mathbf{x}_N(\mathbf{u}_N, \Delta t)$$

$$\mathbf{u}_i = \begin{bmatrix} v_i \\ \omega_i \end{bmatrix}$$

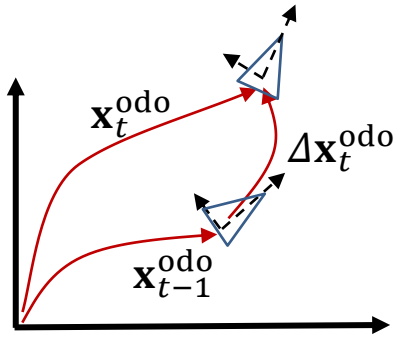
This path results from the **composition of small pose increments** $\Delta\mathbf{x}_i(\mathbf{u}_i, \Delta t)$

Deterministic Motion Models

Odometry-input Motion Model:

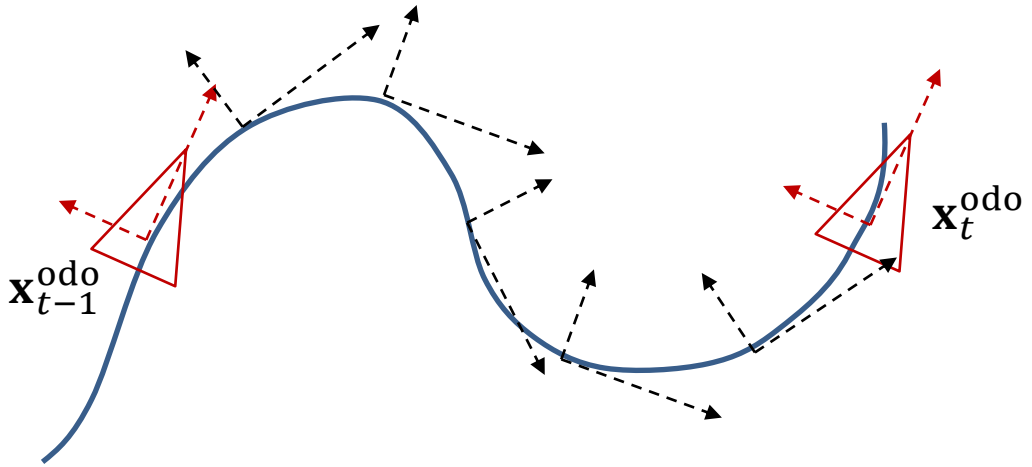


$\Delta\mathbf{x}_t^{\text{odo}} = [\Delta x_t, \Delta y_t, \Delta\theta_t]^{\mathrm{T}}$ *is a* **pose increment** between the poses $\mathbf{x}_{t-1}^{\text{odo}}$ and $\mathbf{x}_t^{\text{odo}}$ provided by the robot's odometry module.



$\Delta\mathbf{x}_t^{\text{odo}}$: computed

$\mathbf{x}_{t-1}^{\text{odo}}$ and $\mathbf{x}_t^{\text{odo}}$: given



$\Delta\mathbf{x}_t^{\text{odo}}$ is expressed in the local odometry frame at $t-1$.

Wheel Encoders for Odometry

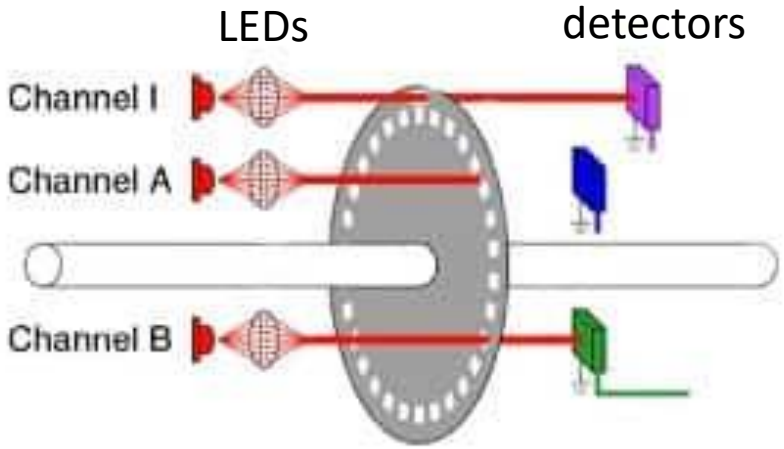
Wheel odometry commonly uses incremental encoders—optical or magnetic—to measure wheel-angle increments.

An optical incremental encoder measures **wheel-angle increments** by detecting light pulses.

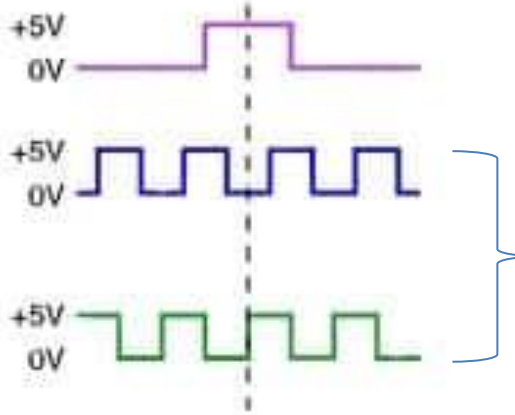
- Light detectors generate pulses as the encoder disk rotates
- Several channels are used to sense rotation



Optical encoder



Example encoder signals



Index pulse: one per revolution

The number of **transitions** gives the angular displacement, while their **temporal order** determines the sign/direction.

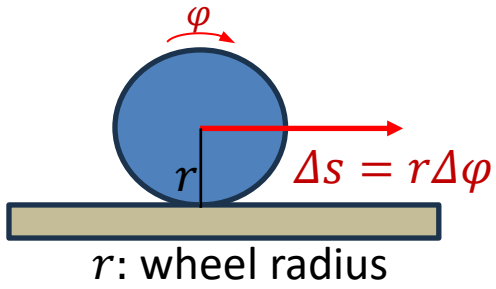
- One direction: 00 → 10 → 11 → 01 → 00
- The other: 00 → 01 → 11 → 10 → 00

Cheap encoders: 2000 – 5000 counts/revolution

Odometry-input Motion Model

What does the wheel-controller (for differential-drive kinematics)?

1. Encoders give the incremental angles $\Delta\varphi_R, \Delta\varphi_L$ of the wheels at intervals δ_t

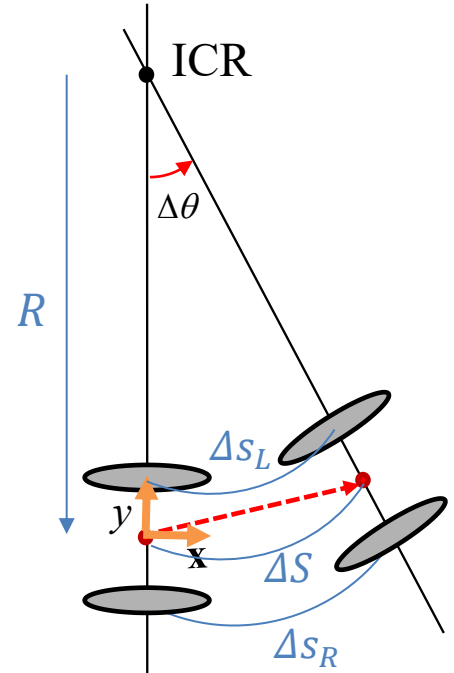


2. Distance traveled by each wheel

$$\Delta s_R = r_R \Delta\varphi_R \qquad \Delta s_L = r_L \Delta\varphi_L$$

3. Assuming constant-curvature, arc and turn traveled by the robot-axis midpoint of the vehicle

$$\Delta s = \frac{\Delta s_R + \Delta s_L}{2} \qquad \Delta\theta = \frac{\Delta s_R - \Delta s_L}{l}$$

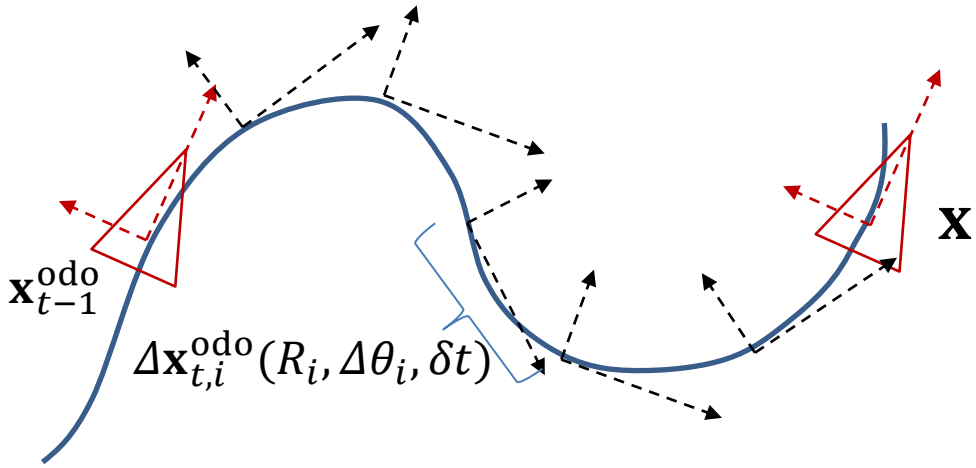


4. Compute incremental coordinates

$$\Delta\mathbf{x} = \begin{bmatrix} \frac{\Delta s}{\Delta\theta}^{R}\sin(\Delta\theta) \\ {}^{R}\frac{\Delta s}{\Delta\theta}[1-\cos(\Delta\theta)] \\ \Delta\theta \end{bmatrix}$$

Odometry Motion Model

The composition of a sequence of $\Delta\mathbf{x}_{t,i}^{\text{odo}}$ between time $t-1$ and t estimates the odometric path between poses $\mathbf{x}_{t-1}^{\text{odo}}$ and $\mathbf{x}_{t}^{\text{odo}}$.



$$\mathbf{x}_{t}^{\text{odo}} = \mathbf{x}_{t-1}^{\text{odo}} \oplus \Delta\mathbf{x}_{t,1}^{\text{odo}} \oplus \cdots \oplus \Delta\mathbf{x}_{t,N_t}^{\text{odo}}$$

published by the controller to the navigation module

N_t: high-rate encoder updates between t−1 and t.

- This is implemented in firmware (running on a robotic platform microcontroller) which sequentially composes pose increments $\Delta\mathbf{x}_i$ at a very high rate, e.g. 100 Hz ($\delta t = 10ms$)
- The odometry poses $\mathbf{x}_t^{\text{odo}} = [x_t^{\text{odo}}, y_t^{\text{odo}}, \theta_t^{\text{odo}}]$ are published to the navigation module at lower rate, e.g. 10 Hz ($\Delta t = 100ms$)

Probabilistic Motion Models

Motion information is imperfect:

- commanded velocities are not executed exactly, and
- odometry increments do not exactly equal the true robot displacement.

Two kinds of errors:

- **Systematic**:
  - introduces a repeatable bias
  - can be modeled or reduced through calibration

Caused by:

- Incorrect or unequal wheel radii
- Incorrect wheelbase
- Actuator bias
- Kinematic-model bias

- **Random**:
  - varies unpredictably between repeated executions
  - is represented by a probability distribution

- Wheel slip
- Terrain irregularities
- Encoder quantization
- External disturbances

We will model this random error next!

Probabilistic Motion Models

Deterministic representation (No error in $\mathbf{u}_t$) -> exact prediction of $\mathbf{x}_t$

$$\mathbf{x}_t = \mathbf{g}(\mathbf{x}_{t-1}, \mathbf{u}_t)$$

Given, input

Probabilistic representation (Error in $\mathbf{u}_t$) -> random prediction of $\mathbf{x}_t$

$$p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t)$$

pose at time t

motion input over [t-1, t]

pose at time t-1

Given, input

This applies to both:

- Velocity motion model: $\mathbf{u}_t = [v_t, \omega_t]^{\mathrm{T}}$
- Odometry motion model: $\mathbf{u}_t = \Delta\mathbf{x}_t^{\mathrm{odo}} = [\Delta x_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$

Review of Probability Fundamentals

Random Variables and Probability Distributions

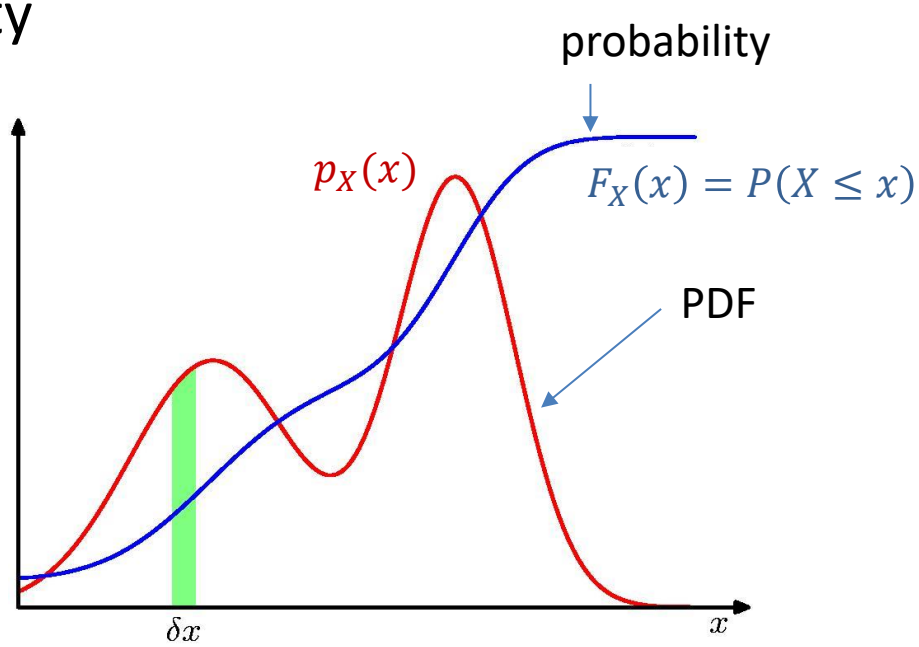
- A random variable X represents an uncertain quantity
- A realization x is one possible value of X
- A continuous random variable is described by a "probability density function" PDF $p_X(x)$

$$p_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} p_X(x)\, dx = 1$$

- From a pdf, a Probability is obtained by integration:

$$P(a < X \leq b) = \int_a^b p_X(x)\, dx$$

Area between a and b



The green area gives a probability:
$P(x_0 < X < x_0 + \delta x) \approx p_X(x_0)\, \delta x$

- For a continuous random variable $P(X = x) = 0$, although $p_X(x)$ need not be zero

Review of Probability Fundamentals

Random Vectors

The robot pose is not a scalar random variable; it is a random vector:

$$\mathbf{X} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \\ \Theta \end{bmatrix} \xrightarrow[\text{possible value}]{\text{particular}} \mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

Notation for simplicity,

- we use lowercase $\mathbf{x}$ for the robot state, whether it is treated as known or uncertain.
- Its meaning will be clear from context.

Probability density for the robot pose at time t:

$$p_{X_t|X_{t-1},U_t}(\mathbf{x}_t|\mathbf{X}_{t-1} = \mathbf{x}_{t-1}, \mathbf{U}_t = \mathbf{u}_t)$$

pose at time t

pose at time t-1

motion input over [t-1, t]

The realization of the variables $\mathbf{X}_{t-1}$ and $\mathbf{U}_t$ are known

Notation for simplicity

$$p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t)$$

Review of Probability Fundamentals

Joint and conditional distributions

Joint density: The joint PDF describes two random variables X and Y simultaneously

$$p_{X,Y}(x,y) = p(x,y)$$

Joint density evaluated at the realization (x, y)

Product Rule: $p(x,y) = p(x|y)p(y) = p(y|x)p(x)$

$$p(y|x) = \frac{p(x,y)}{p(x)}$$

$$p(x|y) = \frac{p(x,y)}{p(y)}$$

Conditional density of X at x, given that $Y = y$

Review of Probability Fundamentals

Marginalization and Total Probability

Marginal density :

From the joint we can

marginalize out y

$$p(x) = \int_y p\,(x, y)dy$$

Sum Rule (Marginalization)

marginalize out x

$$p(y) = \int_x p\,(x, y)dx$$

Combine product rule and marginalization: Law of total probability

$$p(x) = \int_y p\,(x, y)dy = \int_y p(x|y)p(y)dy$$

$$p(x, y) = p(x|y)p(y)$$

Product Rule

Review of Probability Fundamentals

Independence of random variables (RVs)

Definition: $X \perp Y \iff p(x,y) = p(x)p(y)$

Equivalently: $p(x \mid y) = p(x)$

Conditional independence

$X \perp Y|Z$: X conditionally independent of Y **given** Z if

$$p(x,y|z) = p(x|z)p(y|z)$$

At least 3 RVs in play!

That is: $p(x|z) = p(x|z,y)$ and $p(y|z) = p(y|z,x)$

Given z, knowing the values of these RVs has no influence

Review of Probability Fundamentals

Conditional independence is applied in the probabilistic motion model

$$p(\mathbf{x}_t|\underbrace{\mathbf{x}_1, \dots, \mathbf{x}_{t-1}}, \underbrace{\mathbf{u}_1, \dots, \mathbf{u}_t}) = p(\mathbf{x}_t|\mathbf{x}_{1:t-1}, \mathbf{u}_{1:t}) = p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t)$$

All previous poses until $t-1$

All motion commands until t

A more compact notation

Conditional independence

$$\mathbf{X}_t \perp \mathbf{X}_{1:t-2}|\mathbf{x}_{t-1}, \mathbf{u}_t$$

$$\mathbf{X}_t \perp \mathbf{U}_{1:t-1}|\mathbf{x}_{t-1}, \mathbf{u}_t$$

Given the previous pose $\mathbf{x}_{t-1}$ and current motion input $\mathbf{u}_t$, earlier poses and inputs provide no additional information about the next pose.

This is called: Markov assumption

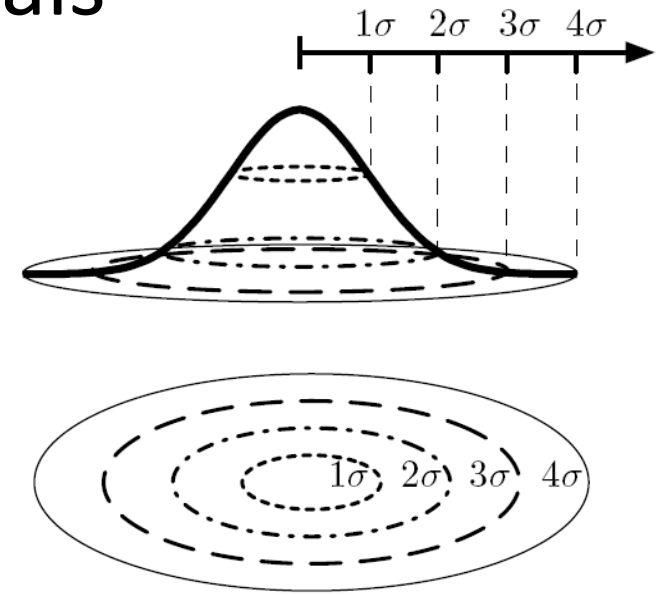
Review of Probability Fundamentals

Gaussian pdf

$$p(\mathbf{x}) = N(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}} \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})}$$

A number that gives the **height** of the bell (for the area to sum 1)

A quadratic polynomial that gives the **position** (μ) and the **shape** (Σ) of the bell



equal-density ellipses

Mean and covariance completely characterize a Gaussian distribution

Mean: $\boldsymbol{\mu} = E[\mathbf{x}] = \begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \\ \boldsymbol{\mu}_\theta \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mu}_r \\ \boldsymbol{\mu}_\theta \end{bmatrix}$ ($\boldsymbol{\mu}_r$)

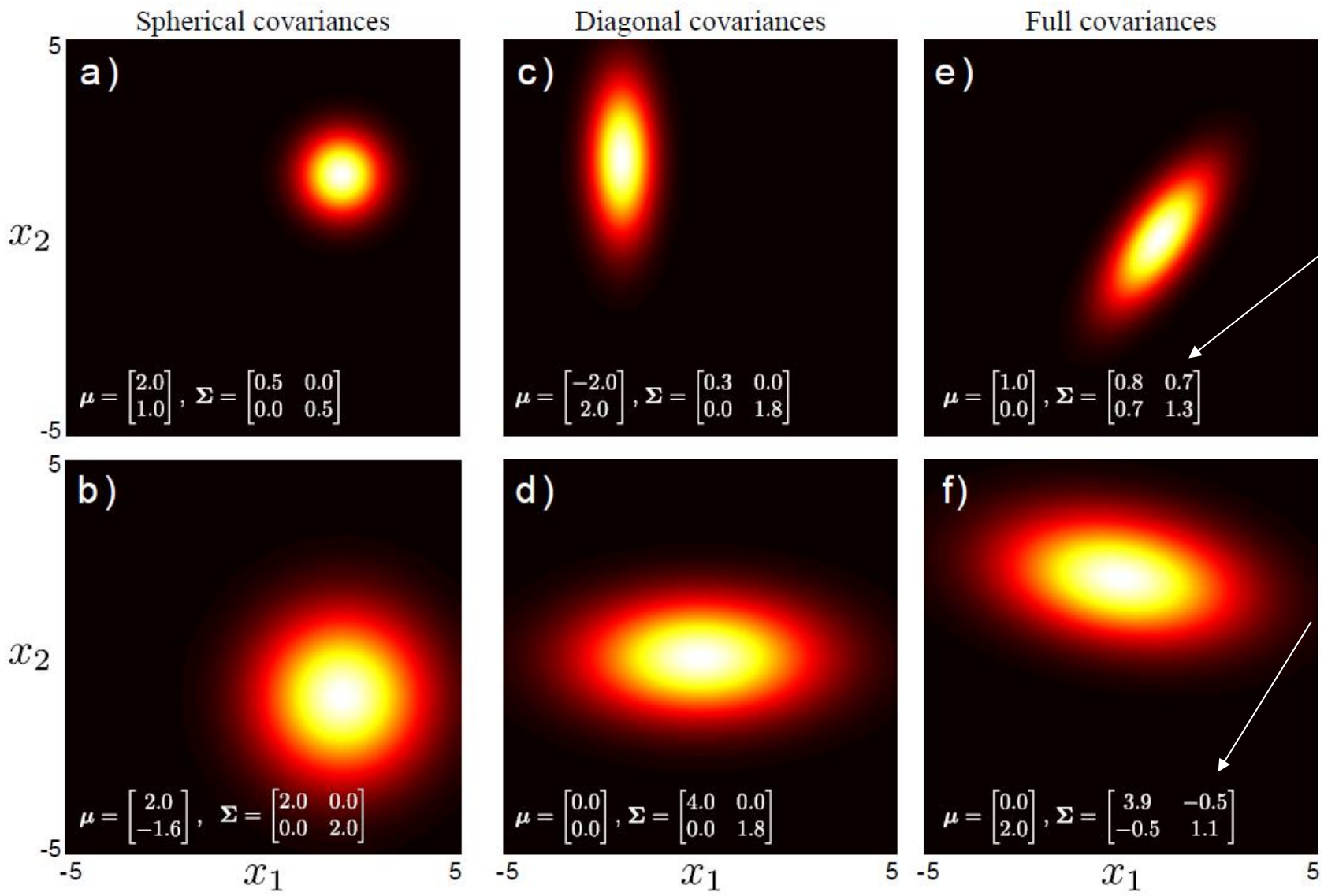
Expectation

Covariance: $\Sigma = \mathrm{cov}[\mathbf{x}] = E[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}] = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{x\theta} \\ \sigma_{xy} & \sigma_y^2 & \sigma_{y\theta} \\ \sigma_{x\theta} & \sigma_{y\theta} & \sigma_\theta^2 \end{bmatrix} = \begin{bmatrix} \Sigma_r & \Sigma_{r\theta} \\ (\Sigma_{r\theta})^{\mathrm{T}} & \Sigma_\theta \end{bmatrix}$ (Σ_r, $\Sigma_{r\theta}$)

Expectation

Off-diagonal elements are cross-covariances; nonzero values indicate linear dependence.

Gaussian pdf



Positive correlation (0.7)

When
$\Delta x = (x - \mu_x) > 0 \rightarrow \Delta y = (y - \mu_y) > 0$

Negative correlation (-0.5)

When $\Delta x > 0 \rightarrow \Delta y < 0$

The covariance matrix determines the shape, size, and orientation of the equal-density ellipses.

Probabilistic Motion Models

Probabilistic representation of poses may come in two forms:

- **Analytic:** Typically, a Gaussian (Normal) distribution

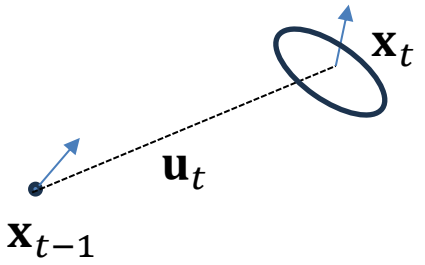
$$\mathbf{x}_t \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t) \approx N(\mathbf{x}_t; \bar{\mathbf{x}}_t, \Sigma_{x_t})$$

"distributed according to"

Random variable vector

Covariance matrix

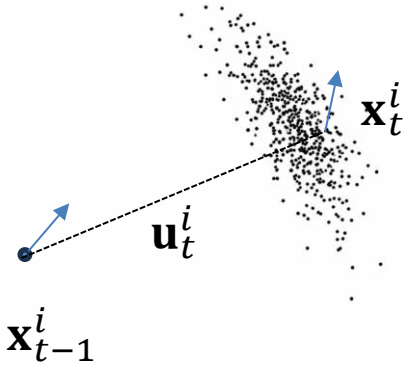
Mean vector



- **Sample:**

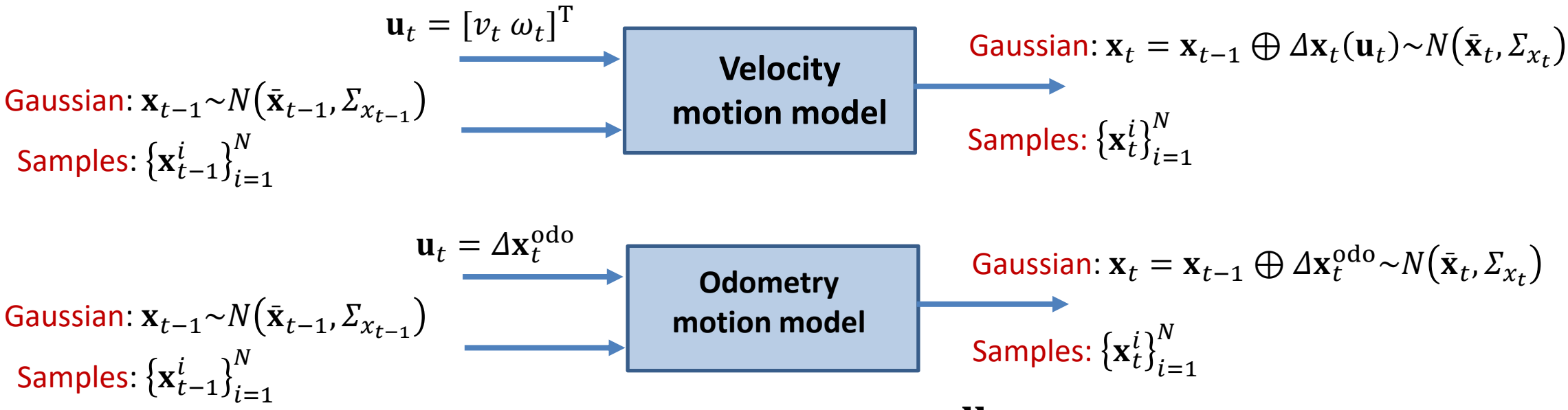
$$\{\mathbf{x}_t^i\}_{i=1}^N \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t)$$

"is a sample drawn from"



Probabilistic Motion Models

Thus, we have 4 possibilities:



$\mathbf{u}_t$

$\mathbf{x}_t$	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t\ \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta x_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\tilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\tilde{u}_t})$	$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

Next, we'll present these 4 cases

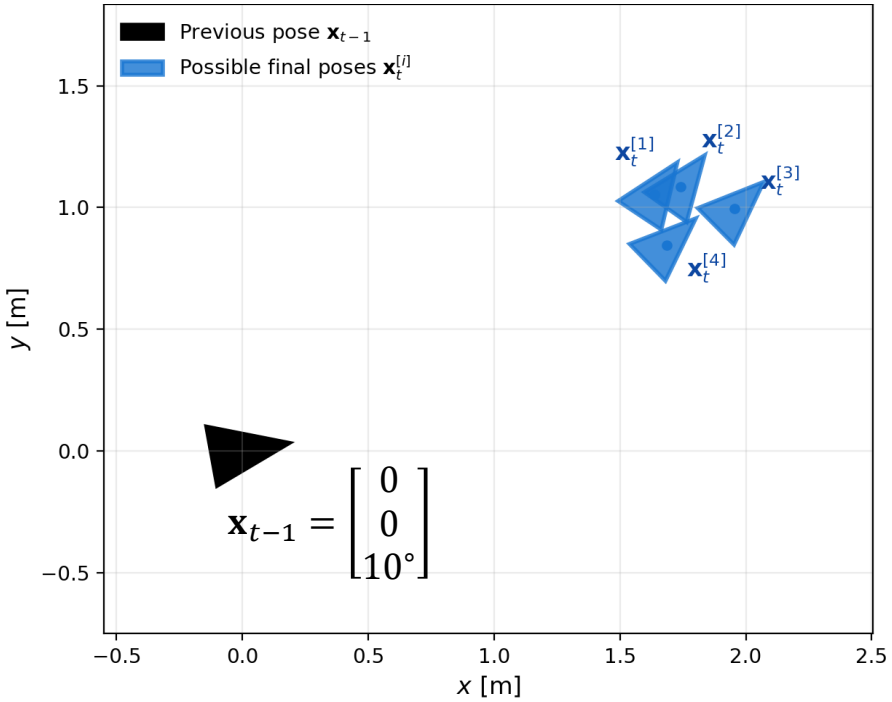
	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t\ \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta \mathrm{x}_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\tilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\tilde{u}_t})$	$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

Velocity-input/Sample-pose. Example

$$\tilde{\mathbf{u}}_t^{[i]} \sim N\left(\mathbf{u}_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}, M_t = \begin{bmatrix} \sigma_v^2 & 0 \\ 0 & \sigma_\omega^2 \end{bmatrix}\right) = N\left(\begin{bmatrix} 1.00 \\ 0.35 \end{bmatrix}, \begin{bmatrix} 0.07^2 & 0 \\ 0 & 0.045^2 \end{bmatrix}\right)$$

velocity command

Repeated execution of the same velocity command $\mathbf{u}_t$ produces different final poses.



4 samples generated from $\tilde{\mathbf{u}}_t \sim N(\mathbf{u}_t, M_t)$

$\tilde{\mathbf{u}}_t^{[1]} = [0.9995, 0.3971]^T$
$\tilde{\mathbf{u}}_t^{[2]} = [1.0519, 0.3826]^T$
$\tilde{\mathbf{u}}_t^{[3]} = [1.1133, 0.2957]^T$
$\tilde{\mathbf{u}}_t^{[4]} = [0.9561, 0.2906]^T$

Apply motion model to each sample

$$\mathbf{x}_t^{[i]} = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t\left(\tilde{\mathbf{u}}_t^{[1]}, \Delta t\right)$$

$\Delta t = 2.0s$

$$\Delta x_t^{[i]} = \frac{\tilde{v}_t^{[i]}}{\tilde{\omega}_t^{[i]}} \sin(\tilde{\omega}_t^{[i]} \Delta t)$$

$$\Delta y_t^{[i]} = \frac{\tilde{v}_t^{[i]}}{\tilde{\omega}_t^{[i]}} \left[1 - \cos(\tilde{\omega}_t^{[i]} \Delta t)\right]$$

$$\Delta\theta_t^{[i]} = \tilde{\omega}_t^{[i]} \Delta t$$

$\mathbf{x}_t^{[1]} = [1.6374, 1.0533, 55.50°]^T$
$\mathbf{x}_t^{[2]} = [1.7425, 1.0854, 53.84°]^T$
$\mathbf{x}_t^{[3]} = [1.9561, 0.9943, 43.89°]^T$
$\mathbf{x}_t^{[4]} = [1.6852, 0.8456, 43.30°]^T$

	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t \; \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta \mathrm{x}_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\tilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\tilde{u}_t})$	$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

Velocity-input/Gaussian-pose

The velocity command $\mathbf{u}_t$ is not executed exactly, but comes with an unbiased error: $\boldsymbol{\xi}_t \sim N(0, M_t)$ with $M_t = \mathrm{Cov}(\boldsymbol{\xi}_t)$

Actual executed velocity: $\tilde{\mathbf{u}}_t = \mathbf{u}_t + \boldsymbol{\xi}_t \rightarrow p(\tilde{\mathbf{u}}_t|\mathbf{u}_t) = N(\tilde{\mathbf{u}}_t; \mathbf{u}_t, M_t)$

Since $\tilde{\mathbf{u}}_t$ is random, the resulting pose $\mathbf{x}_t$ is also random:

$$\mathbf{x}_t = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t(\tilde{\mathbf{u}}_t, \Delta t)$$

What is the distribution of $\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t$?

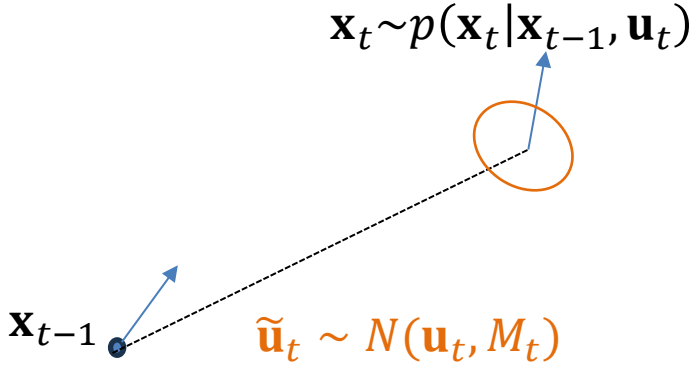
$$p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t) = \int p(\mathbf{x}_t, \tilde{\mathbf{u}}_t|\mathbf{x}_{t-1}, \mathbf{u}_t)\, d\tilde{\mathbf{u}}_t$$

Marginalization

Total probability

$$= \int p(\mathbf{x}_t|\mathbf{x}_{t-1}, \tilde{\mathbf{u}}_t)\, p(\tilde{\mathbf{u}}_t|\mathbf{u}_t)\, d\tilde{\mathbf{u}}_t$$

$p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t)$ will be approximated by a Gaussian pdf (next)



$\mathbf{x}_{t-1}$ is treated as exact (known), **uncertainty comes only from $\tilde{\mathbf{u}}_t$**

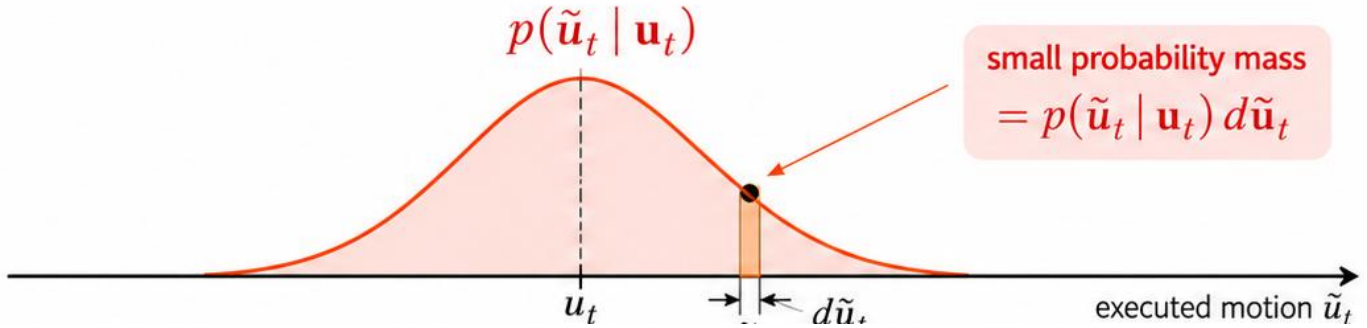
Example 1-D Motion Model (robot in a corridor)

1 Executed motion distribution

The commanded motion u_t is executed with Gaussian error:

$$\tilde{\mathbf{u}}_t = \mathbf{u}_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, M_t)$$

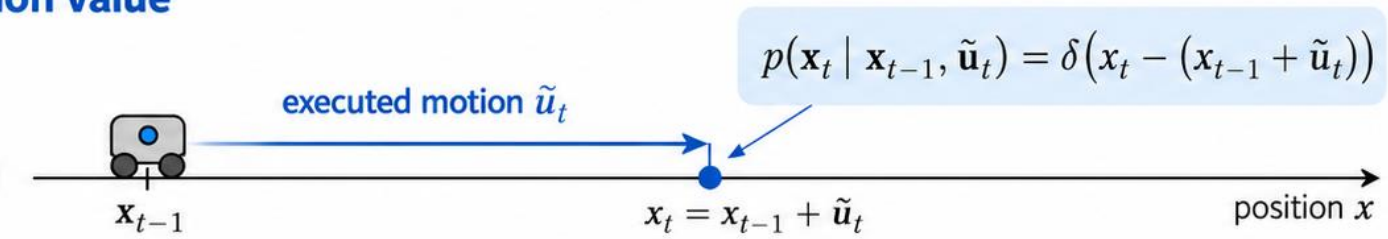
$$\Rightarrow \quad p(\tilde{\mathbf{u}}_t \mid \mathbf{u}_t) = \mathcal{N}(\tilde{\mathbf{u}}_t; \mathbf{u}_t, M_t)$$



2 Contribution of one executed motion value

For a particular executed motion $\tilde{\mathbf{u}}_t$, the resulting position is deterministic:

$$p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \tilde{\mathbf{u}}_t) = \delta\big(x_t - (\mathbf{x}_{t-1} + \tilde{\mathbf{u}}_t)\big)$$

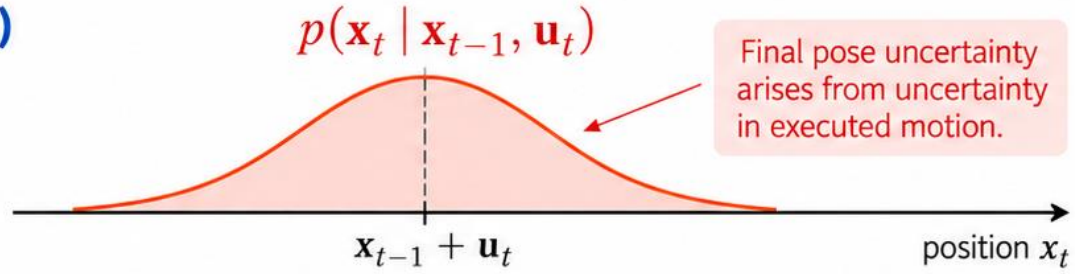


For this small range $d\tilde{\mathbf{u}}_t$, contribution to final pose distribution: $p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \tilde{\mathbf{u}}_t) \cdot p(\tilde{\mathbf{u}}_t \mid \mathbf{u}_t)\, d\tilde{\mathbf{u}}_t = \delta\big(x_t - (x_{t-1} + \tilde{\mathbf{u}}_t)\big) \cdot p(\tilde{\mathbf{u}}_t \mid \mathbf{u}_t)\, d\tilde{\mathbf{u}}_t$

3 Final position distribution (law of total probability)

Add/integrate the weighted contribution of every possible executed motion:

$$p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t) = \int p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \tilde{\mathbf{u}}_t)\, p(\tilde{\mathbf{u}}_t \mid \mathbf{u}_t)\, d\tilde{\mathbf{u}}_t$$



First-order propagation of uncertainty/covariance

Suppose a variable $\mathbf{z}$ as a function of a Gaussian random variable $\mathbf{a}$:

$$\mathbf{z} = \mathbf{f}(\mathbf{a}), \text{ with } \mathbf{a} \sim N(\bar{\mathbf{a}}, \Sigma_a)$$

Because $\mathbf{f}$ may be nonlinear, linearize it around the input means:

$$\mathbf{z} \approx \mathbf{f}(\bar{\mathbf{a}}) + F_a(\mathbf{a} - \bar{\mathbf{a}}), \qquad \text{where } F_a = \frac{\partial \mathbf{f}}{\partial \mathbf{a}}\Big|_{\bar{a}}$$

Jacobian matrix of **f**

Then:

$$\bar{\mathbf{z}} \approx \mathbf{f}(\bar{\mathbf{a}})$$

Mean of **z**

$$\Sigma_z \approx F_a \Sigma_a F_a^{\mathrm{T}}$$

Covariance of **z**

The essential interpretation:

- A Jacobian F_a transforms small input errors ($\Delta\mathbf{a}$) into output errors($\Delta\mathbf{f}$).
- The input covariance Σ_a is transformed into the output space $F_a \Sigma_a F_a^{\mathrm{T}}$.

	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t \; \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta \mathrm{x}_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\tilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\tilde{u}_t})$	$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

Velocity-input/Gaussian-pose

Gaussian approximation of the predicted pose with **known** $\mathbf{x}_{t-1}$

$\mathbf{x}_{t-1}$ is exact (not a random variable)

$$\tilde{\mathbf{u}}_t \sim N(\tilde{\mathbf{u}}_t; \mathbf{u}_t, M_t) \rightarrow \quad M_t = \mathrm{Cov}(\tilde{\mathbf{u}}_t) = \mathrm{Cov}(\boldsymbol{\xi}_t) = \begin{bmatrix} \sigma_v^2 & \sigma_{v\omega} \\ \sigma_{v\omega} & \sigma_\omega^2 \end{bmatrix}$$

Covariance of the velocity uncertainty:

The new pose is:

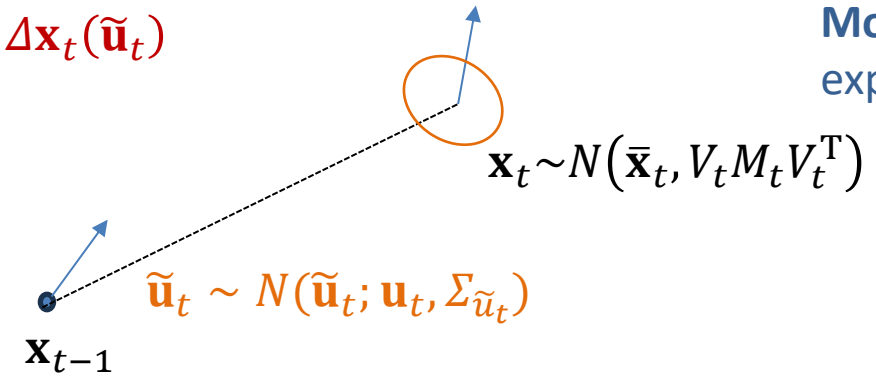
$$\mathbf{x}_t = \mathbf{g}_v(\mathbf{x}_{t-1}, \tilde{\mathbf{u}}_t) \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$$

$$\bar{\mathbf{x}}_t \approx \mathbf{g}_v(\mathbf{x}_{t-1}, \mathbf{u}_t) = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t(\mathbf{u}_t)$$ Predicted mean

$$\Sigma_{x_t} \approx V_t M_t V_t^{\mathrm{T}}$$ Predicted covariance

Non-linear function:
$\mathbf{x}_t = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t(\tilde{\mathbf{u}}_t)$

Motion uncertainty expressed in the pose space



Jacobian matrix:

$$V_t = \left.\frac{\partial \mathbf{g}_v}{\partial \tilde{\mathbf{u}}_t}\right|_{u_t}$$ dim 3x2

Velocity-input/Gaussian-pose

Both pose $\mathbf{x}_{t-1}$ and $\tilde{\mathbf{u}}_t$ are uncertain:

$$\mathbf{x}_{t-1} \sim N\left(\bar{\mathbf{x}}_{t-1}, \Sigma_{x_{t-1}}\right) \quad \tilde{\mathbf{u}}_t \sim N(\mathbf{u}_t, M_t)$$

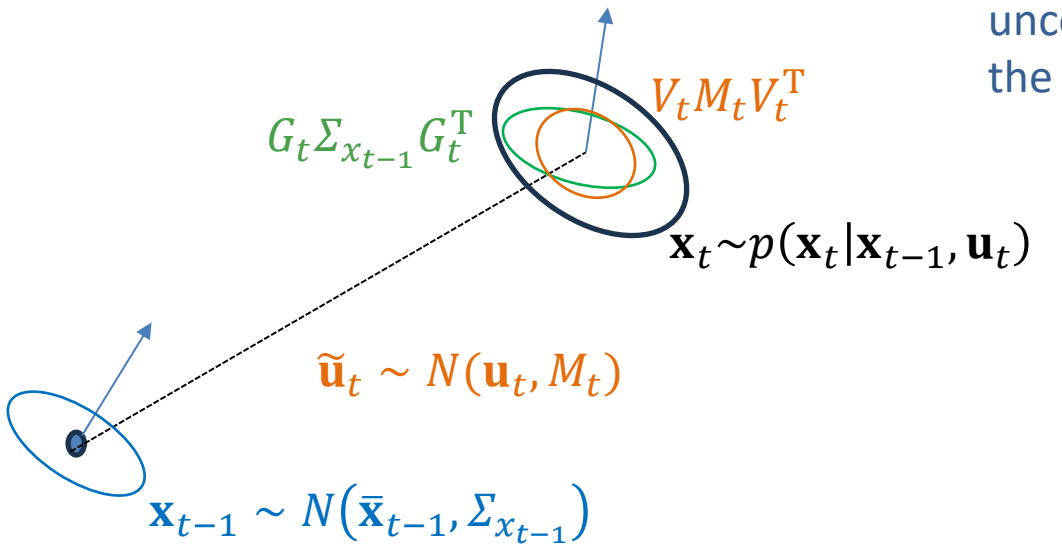
The new robot pose is:

$$\mathbf{x}_t = \mathbf{g}_v(\mathbf{x}_{t-1}, \tilde{\mathbf{u}}_t) \sim N\left(\bar{\mathbf{x}}_t, \Sigma_{x_t}\right)$$

$$\bar{\mathbf{x}}_t \approx \mathbf{g}_v(\bar{\mathbf{x}}_{t-1}, \mathbf{u}_t) \quad \text{Predicted mean}$$

$$\Sigma_{x_t} \approx \underbrace{G_t \Sigma_{x_{t-1}} G_t^{\mathrm{T}}}_{\text{Previous-pose uncertainty in the pose space}} + \underbrace{V_t M_t V_t^{\mathrm{T}}}_{\text{Motion uncertainty expressed in the pose space}} \quad \text{Predicted covariance}$$

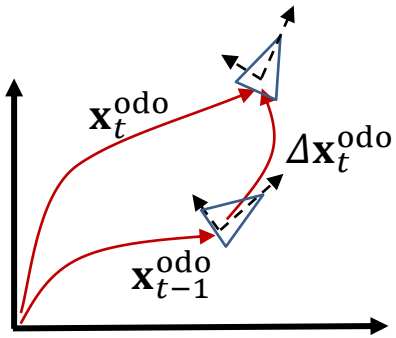
Non-linear function:
$\mathbf{x}_t = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t(\tilde{\mathbf{u}}_t)$



The Jacobians:

$$G_t = \left.\frac{\partial \mathbf{g}_v}{\partial \mathbf{x}_{t-1}}\right|_{\bar{\mathbf{x}}_{t-1}, \mathbf{u}_t} \qquad V_t = \left.\frac{\partial \mathbf{g}_v}{\tilde{\mathbf{u}}_t}\right|_{\bar{\mathbf{x}}_{t-1}, \mathbf{u}_t}$$

	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t \ \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta \mathrm{x}_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\tilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\tilde{u}_t})$	$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$



Odometry-input/Gaussian-pose

We have the readings $\mathbf{x}_{t-1}^{\mathrm{odo}}, \mathbf{x}_t^{\mathrm{odo}}$ and compute $\Delta\bar{\mathbf{x}}_t^{\mathrm{odo}} = \ominus \mathbf{x}_{t-1}^{\mathrm{odo}} \oplus \mathbf{x}_t^{\mathrm{odo}}$

The Gaussian pose $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$ is computed as:

Mean: $\bar{\mathbf{x}}_t = \mathbf{g}_o(\bar{\mathbf{x}}_{t-1}, \Delta\bar{\mathbf{x}}_t^{\mathrm{odo}}) = \bar{\mathbf{x}}_{t-1} \oplus \Delta\bar{\mathbf{x}}_t^{\mathrm{odo}}$

Covariance: $\Sigma_{x_t} \approx \frac{\partial \mathbf{g}_o}{\partial \mathbf{x}_{t-1}} \Sigma_{x_{t-1}} \left[\frac{\partial \mathbf{g}_o}{\partial \mathbf{x}_{t-1}}\right]^{\mathrm{T}} + \frac{\partial \mathbf{g}_o}{\partial \Delta\mathbf{x}_t^{\mathrm{odo}}} \Sigma_{\Delta x_t^{\mathrm{odo}}} \left[\frac{\partial \mathbf{g}_o}{\partial \Delta\mathbf{x}_t^{\mathrm{odo}}}\right]^{\mathrm{T}}$

Covariance at $\mathbf{x}_{t-1}$

No correlation assumed

$$\Sigma_{\Delta x_t^{\mathrm{odo}}} = \begin{bmatrix} \sigma_{\Delta x}^2 & 0 & 0 \\ 0 & \sigma_{\Delta y}^2 & 0 \\ 0 & 0 & \sigma_{\Delta\theta}^2 \end{bmatrix}$$

Jacobians:

$$\frac{\partial \mathbf{g}}{\partial \mathbf{x}_{t-1}} = \begin{bmatrix} 1 & 0 & -\Delta x_t \sin\theta_{t-1} - \Delta y_t \cos\theta_{t-1} \\ 0 & 1 & \Delta x_t \cos\theta_{t-1} - \Delta y_t \sin\theta_{t-1} \\ 0 & 0 & 1 \end{bmatrix} \qquad \frac{\partial \mathbf{g}}{\partial \Delta\mathbf{x}_t^{\mathrm{odo}}} = \begin{bmatrix} \cos\theta_{t-1} & -\sin\theta_{t-1} & 0 \\ \sin\theta_{t-1} & \cos\theta_{t-1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Worked example: Gaussian propagation for odometry-input motion

1. Given data

Given prior pose mean:

$$\bar{\mathbf{x}}_{t-1} = [\,2.00\ \text{m},\ 1.00\ \text{m},\ 30^\circ\,]^{\mathrm{T}}$$

Given prior covariance:

$$\Sigma_{x_{t-1}} = \begin{bmatrix} 0.040 & 0 & 0 \\ 0 & 0.040 & 0 \\ 0 & 0 & 0.0076 \end{bmatrix}$$

Given odometry increment mean:

$$\Delta\bar{\mathbf{x}}_t^{\mathrm{odo}} = [\,1.00\ \text{m},\ 0.20\ \text{m},\ 10^\circ\,]^{\mathrm{T}}$$

Given odometry covariance:

$$\Sigma_{\Delta x_t^{\mathrm{odo}}} = \begin{bmatrix} 0.090 & 0 & 0 \\ 0 & 0.010 & 0 \\ 0 & 0 & 0.0027 \end{bmatrix}$$

Angles are in degrees for readability, angular covariances are in rad^2.

2. Motion model and propagated mean

Odometry-input composition:

$$\bar{\mathbf{x}}_t = \mathbf{g}_o\left(\bar{\mathbf{x}}_{t-1}, \Delta\bar{\mathbf{x}}_t^{\mathrm{odo}}\right) = \bar{\mathbf{x}}_{t-1} \oplus \Delta\bar{\mathbf{x}}_t^{\mathrm{odo}}$$

For 2D pose $[x,\ y,\ \theta]^{\mathrm{T}}$:

$$\begin{aligned} x_t &= x_{t-1} + \Delta x \cos\theta_{t-1} - \Delta y \sin\theta_{t-1} \\ y_t &= y_{t-1} + \Delta x \sin\theta_{t-1} + \Delta y \cos\theta_{t-1} \\ \theta_t &= \theta_{t-1} + \Delta\theta \end{aligned}$$

Numerical evaluation $(\theta_{t-1} = 30^\circ)$:

$$\bar{\mathbf{x}}_t = \begin{bmatrix} 2.00 + 1.00\cos 30^\circ - 0.20\sin 30^\circ \\ 1.00 + 1.00\sin 30^\circ + 0.20\cos 30^\circ \\ 30^\circ + 10^\circ \end{bmatrix} = \begin{bmatrix} 2.766 \\ 1.673 \\ 40^\circ \end{bmatrix}$$

3. Jacobians evaluated at the mean

Jacobian w.r.t. previous pose $\mathbf{x}_{t-1}$:

$$F_x = \frac{\partial \mathbf{g}_o}{\partial \mathbf{x}_{t-1}} = \begin{bmatrix} 1 & 0 & -\Delta x \sin\theta_{t-1} - \Delta y \cos\theta_{t-1} \\ 0 & 1 & \Delta x \cos\theta_{t-1} - \Delta y \sin\theta_{t-1} \\ 0 & 0 & 1 \end{bmatrix}$$

Numerical evaluation $(\theta_{t-1} = 30^\circ,\ \Delta x = 1.00,\ \Delta y = 0.20)$:

$$\boldsymbol{F}_x = \begin{bmatrix} 1 & 0 & -0.673 \\ 0 & 1 & 0.766 \\ 0 & 0 & 1 \end{bmatrix}$$

Jacobian w.r.t. odometry increment $\Delta\mathbf{x}_t^{\mathrm{odo}}$:

$$F_u = \frac{\partial \mathbf{g}_o}{\partial \Delta\mathbf{x}_t^{\mathrm{odo}}} = \begin{bmatrix} \cos\theta_{t-1} & -\sin\theta_{t-1} & 0 \\ \sin\theta_{t-1} & \cos\theta_{t-1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Numerical evaluation $(\theta_{t-1} = 30^\circ)$:

$$\boldsymbol{F}_u = \begin{bmatrix} 0.866 & -0.500 & 0 \\ 0.500 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Covariance propagation

First-order propagation (independent uncertainties):

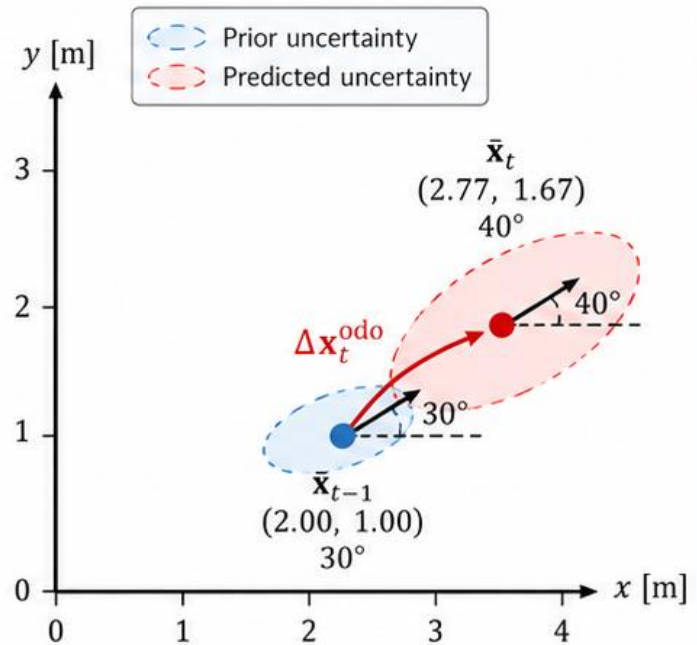
$$\Sigma_{x_t} \approx F_x \Sigma_{x_{t-1}} F_x^{\mathrm{T}} + F_u \Sigma_{\Delta x_t^{\mathrm{odo}}} F_u^{\mathrm{T}}$$

Numerical result:

$$\Sigma_{x_t} \approx \begin{bmatrix} 0.113 & 0.031 & -0.005 \\ 0.031 & 0.074 & 0.006 \\ -0.005 & 0.006 & 0.010 \end{bmatrix}$$

Entries mix m^2, m·rad, and rad^2.

5. Geometric interpretation



Takeaways

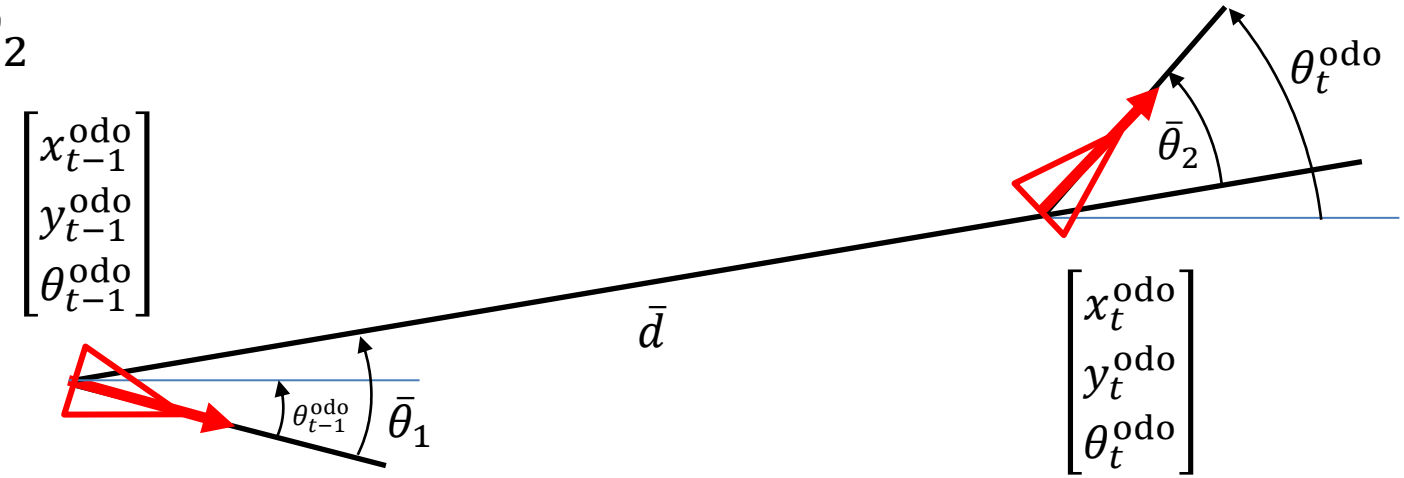
- The mean pose is obtained by composing the previous pose with the odometry increment.
- The Jacobians map small pose and odometry errors into output-pose errors.
- Because the two uncertainty sources are assumed independent, their covariance contributions add.

	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t \ \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta \mathrm{x}_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\widetilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\widetilde{u}_t})$	$\widetilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

Odometry-input/Sample-pose

Again, we have the readings $\mathbf{x}_{t-1}^{\mathrm{odo}}, \mathbf{x}_t^{\mathrm{odo}}$ and now compute the displacement $\mathbf{u}_t = [\bar{\theta}_1, \bar{d}, \bar{\theta}_2]^{\mathrm{T}}$:

The odometry increment is represented by an equivalent rotation θ_1, translation d and rotation θ_2



Rotation at *t-1*: $\bar{\theta}_1 = \mathrm{atan2}(y_t^{\mathrm{odo}} - y_{t-1}^{\mathrm{odo}}, x_t^{\mathrm{odo}} - x_{t-1}^{\mathrm{odo}}) - \theta_{t-1}^{\mathrm{odo}}$

Straight motion at *t*–1: $d = \sqrt{(x_t^{\mathrm{odo}} - x_{t-1}^{\mathrm{odo}})^2 + (y_t^{\mathrm{odo}} - y_{t-1}^{\mathrm{odo}})^2}$

Rotation at *t*: $\bar{\theta}_2 = \theta_t^{\mathrm{odo}} - \theta_{t-1}^{\mathrm{odo}} - \bar{\theta}_1$

	Velocity-input	Odometry-input
Gaussian form $\mathbf{x}_t \sim N(\bar{\mathbf{x}}_t, \Sigma_{x_t})$	$\mathbf{u}_t = [v_t \; \omega_t]^{\mathrm{T}}$	$\mathbf{u}_t = [\Delta \mathrm{x}_t^{\mathrm{odo}}, \Delta y_t^{\mathrm{odo}}, \Delta\theta_t^{\mathrm{odo}}]^{\mathrm{T}}$
Sample form $\{\mathbf{x}_{t-1}^i\}_{i=1}^N$	$\tilde{\mathbf{u}}_t^{[i]} \sim N(\mathbf{u}_t, \Sigma_{\tilde{u}_t})$	$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

Each sample pose $\mathbf{x}_{t-1}^{[i]}$ is moved with a random motion:

$$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}} \sim N\left(\mathbf{u}_t^{\mathrm{odo}}, \Sigma_{\tilde{u}_t^{\mathrm{odo}},}\right)$$

We skip the superscript $[i]$ for clarity

$$\theta_1 \sim N(\bar{\theta}_1, \sigma_{\theta_1}^2)$$
$$d \sim N(\bar{d}, \sigma_d^2)$$
$$\theta_2 \sim N(\bar{\theta}_2, \sigma_{\theta_2}^2)$$

$$\Sigma_{\tilde{u}_t^{\mathrm{odo}}} = \begin{bmatrix} \sigma_{\theta_1}^2 & 0 & 0 \\ 0 & \sigma_d^2 & 0 \\ 0 & 0 & \sigma_{\theta_2}^2 \end{bmatrix}$$

$\tilde{\mathbf{u}}_t^{\mathrm{odo},[i]} = [\theta_1^{[i]}, d^{[i]}, \theta_2^{[i]}]^{\mathrm{T}}$

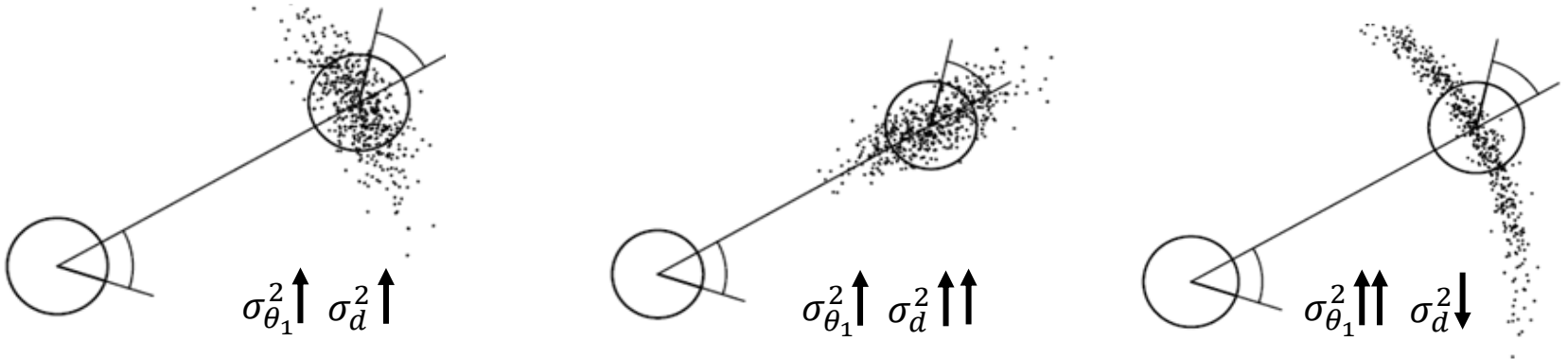
$\{\mathbf{x}_{t-1}^{[i]}\}_{i=1}^N$

Odometry motion model

$\{\mathbf{x}_t^{[i]}\}_{i=1}^N$

$$x_t^{[i]} = x_{t-1}^{[i]} + d^{[i]} \cos(\theta_{t-1}^{[i]} + \theta_1)$$
$$y_t^{[i]} = y_{t-1}^{[i]} + d^{[i]} \sin(\theta_{t-1} + \theta_1)$$
$$\theta_t^{[i]} = \theta_{t-1}^{[i]} + \theta_1 + \theta_2$$

Running this method with different parameter settings. Only showing the positional components.



From: Probabilistic Robotics. Thrun et al. 2005.

Summary

Vehicle kinematics :

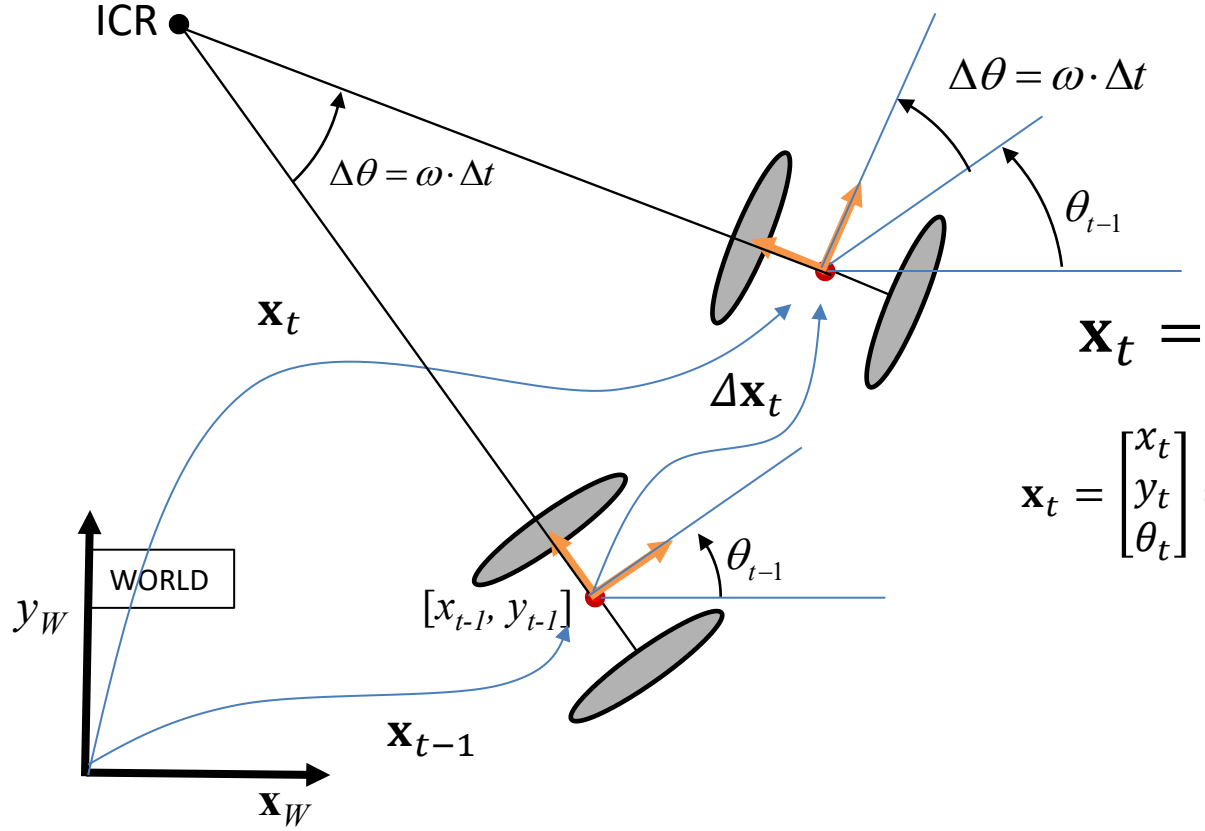
$\mathbf{u} = [v_P, \omega]^{\mathrm{T}}$

or

$\mathbf{u} = [\omega_l, \omega_r]^{\mathrm{T}}$

Robot Kinematics

$$\Delta x = \frac{v_P}{\omega}\sin(\omega \Delta t)$$
$$\Delta y = \frac{v_P}{\omega}[1 - \cos(\omega \Delta t)]$$
$$\Delta\theta = \omega \Delta t$$



$$\mathbf{x}_t = \mathbf{g}(\mathbf{x}_{t-1}, \Delta\mathbf{x}_t) = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t$$

$$\mathbf{x}_t = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} = \begin{bmatrix} x_{t-1} + \Delta x_t \cos\theta_{t-1} - \Delta y_t \sin\theta_{t-1} \\ y_{t-1} + \Delta x_t \sin\theta_{t-1} + \Delta y_t \cos\theta_{t-1} \\ \theta_{t-1} + \Delta\theta_t \end{bmatrix} = \mathbf{x}_{t-1} \oplus \Delta\mathbf{x}_t$$

pose composition

Summary

Deterministic representation (No error in $\mathbf{u}_t$) -> exact prediction of $\mathbf{x}_t$

$$\mathbf{x}_t = \mathbf{g}(\mathbf{x}_{t-1}, \mathbf{u}_t)$$

Probabilistic representation (Error in $\mathbf{u}_t$) -> random prediction of $\mathbf{x}_t$

$$p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t)$$

